∞-Categories

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Certificate

This is to certify that this dissertation entitled ∞-Categories towards the partial fulfilment of the BS-MS dual degree programme at the Indian Institute of Science Education and Research, Pune represents study/work carried out by Dipankar Maity at Indian Institute of Science Education and Research under the supervision of Dr. Amit Hogadi, Associate Professor, Department of Mathematics, during the academic year 2021-2022.

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I dedicate my thesis to my Mom and Briti.

Declaration

I hereby declare that the matter embodied in the report entitled ∞-Categories are the results of the work carried out by me at the Department of Mathematics, Indian Institute of Science Education and Research, Pune, under the supervision of Dr. Amit Hogadi and the same has not been submitted elsewhere for any other degree.

Dipankar Maity

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Abstract

This text attempts to review and understand the fundamentals of ∞-categories via weak Kan complexes. We develop to some extent the construction of ∞ homs. We find a way to deal with 1-morphisms and equivalences and form the ∞-category of all ∞-categories. Finally we present J-model structure on simplicial sets, which represents ∞-categories as fibrant objects. At the end we briefly touch upon further concepts which are possible to be constructed in the theory, in correspondence with ordinary category theory.

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Introduction

Historically the idea that two objects are 'equivalent' up to some condition has been existing since the development of homotopy theory of topological spaces or of homological algebra of R-Modules. But they were just being studied by identifying such objects up to the corresponding 'equivalences'. But since many theories in mathematics such as the theory of stacks or that of 2-categories in general or spectral theory in topology and many other of this type has suggested that the concepts of such 'equivalences' are not that redundant to be discarded right away by taking equivalences classes. In fact, much more information are encoded in the equivalences, for example they have equivalences among themselves, providing a huge set of such equivalence data. It wont be surprising if such data reveal some new and fundamental results in the corresponding theory.

If this follows coherently, one might wonder how such 'equivalences', "equivalences between equivalences" and so on fit together to be described nicely in mathematical words. One such model in nice topological setting (For e.g. CW complexes) is: on degree 0 you put topological spaces, on degree 1 you take continuous maps, on degree 2 homotopies of continuous maps, on degree 3 homotopies between homotopies and so on. It was Grothendieck's philosophy that whatever we were asking about higher equivalences must be reformulable as the situation we just mentioned. Moreover there are two extrema to this question: first, where all morphisms above degree 0 are 'invertible' and second, where none of the morphisms are 'invertible'. The answers were clear: the first one is the fundamental groupoid of a topological space whereas for the second one is a category. Therefore, whatever we seek for should have both topological and categorical nature. It was later understood that simplicial sets are really a good generalization of nice topological spaces as categorical objects, at least from computational aspects. Therefore one might wonder whether Grothendieck's philosophy will still be valid in this case, and it turned out that it is in fact one of the best

fits for such a theory. The whole text we present hereafter will be an elaboration of this idea.

The modern definition as introduced by Joyal[1] and Lurie [2] indicate how the idea of filler along inner horns in simplicial sets has really the power to replace the idea of both composition (filling along inner 2 horn) as well as associativity (filling along inner 3 horns) or whatever else is needed for the higher order compatibilities, which are really hard to describe in a general n-category. In fact, such compatibility laws become too complicated for even $n = 3$ case. Therefore it seems that '∞-categoriesáre easier than n-categories at least in terms of descriptions.

The name ∞-category comes from Lurie's texts. Earlier they were called weak kan complexes in [3] and quasi categories in [1]. However it turns out that if we choose this definition then all the 'morphisms' for degree $n > 1$ become 'invertible' to some sense in nature. Therefore this text is a model for only $(\infty, 1)$-categories, not (∞, n) categories in general.

There are other different models of $(\infty, 1)$ available in the literature. Namely, *Segal categories* [4], *Complete Segal spaces* [5], 'topologically enriched categories', *1-Complicial sets* and popularly *Simplicially enriched categories* [6]. All of them have their own advantages and disadvantages. Fortunately they are all simplicially enriched to some sense with some additional things called *'sketches'* taken care of and thus each of them can easily be accessed whenever needed. We call each of them as ∞-cosmoi and all of them together as an ∞-Cosmos.

Original contributions:

Everything presented in this thesis is original to the best of my knowledge, except for a few technical results, the references for which are mentioned in the respective places. The proof of J-model structure using Jeff smith theorem is entirely new.

Prerequisites

- **Basic Category Theory**
  The theory we present is a generalization of the basic theory of Categories where we have a set of objects and a set of morphisms with some structures among them. Since its a text on ∞-category , the reader should be familiar with basic category theoretic language. Basic results on adjoint functors, Enriched categories and limits/colimits are assumed.

- **Simplicial sets**
  Basic familiarity with simplicial sets is assumed. Furthermore the Quillen injective model structure on simplicial sets plays a crucial role at the core of the theory. Fibrations are kan fibrations and weak equivalences are weak homotopy equivalences.

- **Model categories**
  The concept of Model categories, in particular, simplicial model categories are at the core of the theory. In particular, for the 4th chapter, the notion of combinatorial model structure as introduced by Jeff smith is assumed.

.

Setup and notations

Before we start let us prepare the setups and notations for the text. Throughout this literature we fix a sequence of Grothendieck universes $\mathcal{G}_0 \in \mathcal{G}_1 \in \mathcal{G}_2$. The objects of $\mathcal{G}_0$ are called small sets and we denote it by $Sets$. The objects of $\mathcal{G}_1$ are called large sets and we denote this as $SETs$. And finally the last one has extra large sets and would denote

it by $SETS$. The simplicial sets we consider here are mostly presheaves of $Sets$ on Δ and every ∞-category we define are 'small' ones. However the ∞-category of all ∞-categories is an ∞-category in the next Grothendieck universe $\mathcal{G}_1$. Therefore whatever happens in 'strengthening unstrengthening', really happens in $\mathcal{G}_2$. The extra large class won't arise for us but will always be a parent to hold all the constructions. Unless otherwise stated all ordinary categories will be locally small i.e. the homs are small sets ($\in \mathcal{G}_0$).

We will denote ordinary n-categories (for $n < \omega$) by script letters $\mathscr{A}, \mathscr{B}, \mathscr{C}, ..etc$, and ∞-categories by calligraphy letters $\mathcal{A}, \mathcal{B}, \mathcal{C}, ...etc$, and all higher functors by $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, ..etc$ and 1-functors simply by italic letters $A, B, C..., a, b, c...etc$ or Greek letters. General simplicial sets will be written in bold letters, or occasionally by italic letters.

A simplicial set is a contravarient functor $\Delta^{op} \xrightarrow{\mathbf{B}} Sets$. More concretely it is given by sets $\mathbf{X}_n$ for $n \geq 0$ with morphisms $\mathbf{X}_n \xrightarrow{s_j} \mathbf{X}_{n+1}$ (called degeneracies) for $j \leq n$, and $\mathbf{X}_n \xrightarrow{d_i} \mathbf{X}_{n-1}$ (called faces) for $i \leq n$, satisfying the following conditions:

$$\begin{cases} d_i d_j = d_{j-1} d_i, d_i s_j = s_{j-1} d_i, & \text{if } i < j \\ d_j s_j = 1 = d_{j+1} s_j, & \\ d_i s_j = s_j d_{i-1}, & \text{if } i > j+1 \\ s_i s_j = s_{j+1} s_i, & \text{if} i \leq j \end{cases}$$

By the standard n-simplex Δ^n, we mean the simplicial set represented by $[n]$; by boundary $\partial\Delta^n$ we mean the subsimplex of Δ^n which contains all the faces of the standard n-simplex; and by horn Λ^n_j, we mean smallest the subsimplex which contain all the faces except for the jth one. For a detailed understanding of simplicial sets, one should refer to [7].

The category of (small) simplicial sets will be denoted $sSets$. Whereas the category of (small) categories will be denoted as Cat. Small categories will be $\mathcal{G}_0$-categories and large categories will be $\mathcal{G}_1$-categories. Categories, which are locally small, but skeletally large, will be considered as $\mathcal{G}_1$-categories only (we won't be bothering ourselves by mentioning this every time).

Chapter 1

Composers and ∞ -Categories

In this chapter we introduce the basic notion for composers starting with small ones.After we have done that we will investigate carefully the case of strong ones and come to the description of the simplicity of compatibility for the case of strong composers. We will then define an $\infty - category$ and show how how an $n - category$ can be thought of as an $\infty - category$. The version of ∞-categories we use are also known as *quasicategories* in the literature. There are many other different approaches to this, namely via '*Segal categories*', '*Simplically enriched categories.*' However quasicategories seems to be advantageous among all such. We begin with the following definition.

1.1 Composers

Definition 1.1.1. *A simplicial set* $\mathbf{X}$ *is called a small composer if it has right lifting property along the inner 2-horn inclusion (* $\Lambda^2_1 \hookrightarrow \Delta^2$ *) i.e. the following extension exists.*

$$\begin{tikzcd} \Lambda^2_1 \arrow[r] \arrow[d, hook] & \mathbf{X} \\ \Delta^2 \arrow[ur, dotted] & \end{tikzcd}$$

Clearly one such map from Λ^2_0 to $\mathbf{X}$ is a pair of morphisms $x \xrightarrow{f} y$ and $y \xrightarrow{g} z$.

Such an extension when restricted to $\Delta^{\{0,2\}}$ can be thought of as a composition of g and f. Of course there are many such compositions possible. We will show later that there is a 'simplicial set' of ways to such compositions.

Well now after compositions are defined the next question one should talk about is of course associativity. We will do that in the following way.

Definition 1.1.2. *Let* $\mathbf{X} \in \mathbf{sSet}$ *be a simplicial set and* f, g *be two morphism* $x \underset{g}{\overset{f}{\rightrightarrows}} y$ *. We say* f *is equivalent to* g *if* $\exists \sigma : \Delta^2 \to \mathbf{X}$ *a two simplex such that*

$$\sigma_{|\Delta^{\{0,1\}}} = f \tag{1.1}$$

$$\sigma_{|\Delta^{\{0,2\}}} = g \tag{1.2}$$

$$\sigma_{|\Delta^{\{1,2\}}} = 1_y \tag{1.3}$$

Clearly if $\mathbf{X}$ is a small composer then it is same as saying that g is a composition of 1_y and f.

Proposition 1.1.1. *Let* $\mathbf{X}$ *be a small composer which has in addition the right lifting property with respect to inner 3 horn inclusion (* $\Lambda^3_i \hookrightarrow \Delta^3$ *,* $\forall i$ *s.t.* $0 < i < 3$*). Then any two composition are equivalent. Moreover associativity holds up to equivalence. In fact, the notion of being equivalence, as defined above, is indeed an equivalence relation.*

Proof. Consider the following diagram:

$$\begin{tikzcd}
 & & z & \\
 & & z \arrow[u, "1_z"'] & \\
x \arrow[rr, "f"'] \arrow[urr, "(gf)_2"] \arrow[ur, "(gf)_1"'] & & y \arrow[uu, two heads] \arrow[ul, "g"'] \arrow[uul, "g"] &
\end{tikzcd}$$

Where the bottom face defines $(gf)_1$ as one composite and the front face defines one another composite $(gf)_2$, the t right side behind face is the degenerate 2 simplex on g. Therefore we have the horn Λ^3_1 on the vertex y. Since $\mathbf{X}$ has lifting along inner 3 horns, the above diagram can be completed to Δ^3 Which when restricted to the face $\Delta^{\{0,2,3\}}$ implies that $(gf)_1 \sim (gf)_2$. Whereas the following diagram implies associativity up to equivalence with

the same argument.

$$\begin{tikzcd}
 & w & \\
 & z \arrow[u, "h"'] & \\
x \arrow[ur, "gf"'] \arrow[uur, bend left, "((hg)f"] \arrow[rr, "f"'] & & y \arrow[ul, "g"] \arrow[uul, bend right, "hg"']
\end{tikzcd}$$

$\square$

We would like to call such composers as strong ones. Now one would like to define the composition of n many morphisms unambiguously in general, even when such 3 horn lifting properties are not provided. We will do that in the following way.

Definition 1.1.3. *An n- composer is a Simplicial set which has right lifting property along n spine inclusion (i.e. $I^n \hookrightarrow \Delta^n$). A simplicial set $\mathbf{X}$ is called a composer if it is an n-composer for all n.*

Note that since $I^2 = \Lambda^2_1$, a small composer is just a 2-composer. Intuitively an infinity category should be a replacement of an n-category and hence one should be able to compose any number of morphisms which is the case for composers. However as the above proposition suggests that to deal with associativity problem at each degree one should have the inner horn lifting for each $n \geq 2$. This observation leads to the following section.

1.2 ∞ -Categories

Definition 1.2.1. *An $\infty -$ category is a Simplicial set which has right lifting property with respect to all inner horn inclusions (i.e. $\Lambda^n_j \hookrightarrow \Delta^n$, $\forall\ 0 < j < n$).*

We will see later that having inner horn lifting property guarantee, lifting along spine inclusions, i.e. every $\infty -$ *category* is a composer so it is enough to define an $\infty -$ *category* this way. Once again such lifts are not required to be unique so that everything is defined up to some form of equivalances. The next proposition shows that requiring unique lifts will give us the notion of 1-categories as results, suggesting that we should refrain ourselves from doing that.

Proposition 1.2.1. *Let* $\mathbf{X}$ *be a simplicial set. Then the following are equivalent.*

1. $\mathbf{X}$ *has unique lift along the inner horn inclusions.*

2. $\mathbf{X}$ *has unique lift along spine inclusions.*

3. $\mathbf{X}$ *is the nerve of an ordinary category.*

Proof. The equivalence of 2 and 3 is clear, since in $\mathbf{X} = N(\mathscr{C})$ for some ordinary category $\mathscr{C}$ then any simplice is nothing but a commutative diagram in it and therefore to lift a n spine is just filling up the corresponding edges using compositions and this is the only way as compositions in a category is unique and this proves $3 \implies 2$. Conversely If the category has a unique lift along any spine then one can see that composition as is defined above is unique and as well as the unique 3 spine lifting guarantees associativity and hence gives an ordinary category. The fact that all the higher cells are just given by a path of n morphisms is obvious by uniqueness and hence $2 \implies 3$.

The proof of $3 \implies 1$ would follow from our next concept that the nerve functor N from the category of small categories to the category of simplicial sets has a left adjoint $\mathbf{h}$. We have given an explicit proof of this fact in section . Now since $\Lambda^2_0 = \mathbf{I}^2$ $1 \implies 2$ or $n = 2$. For $n = 3$ given a map from spine to $\mathbf{X}$ one can fill up each of the faces except $\Delta^{\{0,2,3\}}$ as once again $\Lambda^2_0 = \mathbf{I}^2$ and thus get the horn Λ^3_1 which can be lifted uniquely by 1 and thus giving a lift for the spine inclusion which is again unique as on faces they restricts to a unique lifts. The general case follows inductively. $\square$

Definition 1.2.2. *Given a simplicial set* $\mathbf{X}$ *the homotopy category* $h(\mathbf{X})$ *is defined to be the following category.*

1. *For class of objects take* $ob(h(\mathbf{X})) = X_0$.

2. *For morphisms take* $h(\mathbf{X})(x, y)$ *to be the equivalence class of free concatination of 1 simplicies starting at* x *and ending at* y *. Where two such paths are said to be equivalent if if in any higher simplex both of them lie in the 1 skeleton with* x *at the* 0*th vertex and the end points of them are connected by* $s_0(y)$ *. Of course this is not a equivalence relation and one has to generate one such out of it.*

Its quite clear that if $\mathbf{X}$ is a strong composer then any path of morphisms is replaceable by a single morphism which is determined up to equivalence and hence $h(\mathbf{X})(x,y) = \mathbf{X}_1(x,y)/\sim$. h thus defined as above on simplicial sets is a functor from from the category of **sSets** Simplicial sets to the category **Cat** of small categories.

Proposition 1.2.2. *There is an adjunction $h \dashv \mathbf{N}$.*

$$h : \mathbf{sSet} \to \mathbf{Cat} : \mathbf{N} \tag{1.4}$$

Proof. The proof is elementary.

Note that the counit of the adjunction is just the identity natural transformation and therefore $\mathbf{N}$ is full and faithful. Since h is a construction involving only 2- skeleton we have that if $sk_2(\mathbf{X}) = sk_2(\mathbf{Y})$ for two simplicial sets then we have that $h(\mathbf{X} = h((Y)))$.

□

Lemma 1.2.3. *Let $n \geq 4$. Then for any $0 \leq j \leq n$ the following lifting problem solves uniquely.*

$$\begin{tikzcd} \Lambda^n_j \arrow[r] \arrow[d, hook] & \mathbf{N}(\mathcal{C}) \\ \Delta^n \arrow[ur, dotted] & \end{tikzcd}$$

Proof. By adjunction the above lifting problem is equivalent to the following lifting problem in **Cat** .

$$\begin{tikzcd} h(\Lambda^n_j) \arrow[r] \arrow[d, hook] & \mathcal{C} \\ h(\Delta^n) \arrow[ur, dotted] & \end{tikzcd}$$

But for $n \geq 4$ we have $sk_2(\Lambda^n_j) = sk_2(\Delta^n)$ we have $h(\Lambda^n_j) \cong h(\Delta^n)$ and hence the last lifting problem solves immediately. □

Proposition 1.2.4. *Nerve of a category has a unique lift along inner horn inclusions.*

Proof. Due to the above proposition it is enough to show the case for $n = 2, 3$. Since $\mathbf{I}^2 = \Lambda^2_1$ the case $n = 2$ is trivial, any such is lift has to be given by the composition only. Now

consider a horn Λ^3_1 in a category as follows:

$$\begin{tikzcd}
 & D & \\
 & C \arrow[u, "h"] & \\
A \arrow[ur, "gf"] \arrow[uur, "((hg)f"] \arrow[rr, "f"'] & & B \arrow[ul, "g"'] \arrow[uul, "hg"']
\end{tikzcd}$$

where the B represents the $1st$ vertex. To say that its the horn Λ^3_1 is to say that all the faces except the left behind face (C is viewed at the back) are commutative. But since in a category composition is associative i.e. $h(gf) = (hg)f$, the left behind face is also commutative and hence the lift exists which is obviously unique. $\square$

Proposition 1.2.5. *Nerve of a category is a kan complex iff the category is a groupoid.*

Proof. Let $\mathcal{C}$ be a groupoid. To show that $N(\mathcal{C})$ is a kan complex we need to solve only the 2 and 3 horn lifting problems by the above lemma. Now since nerves have inner horn lifting property it remains to deal with the case of outer 2 and 3 horns. A map $\Lambda^2_0 \to N(\mathcal{C})$ is given by 2 maps (f, g) where f is the restriction to $\Delta^{\{0,1\}}$ and g is the restriction to $\Delta^{\{0,2\}}$. Since $\mathcal{C}$ is a groupoid f^{-1} exists and a lift is then given by gf^{-1}. The case of the right outer 2 horn follows similarly. Now consider the case of left 3 horn inclusions. A map $\Lambda^3_0 \to N(\mathcal{C})$ is just a map $\Lambda^3_1 \to N(\mathcal{C})$ where one replaces the $\Delta^{\{0,1\}}$ edge by its inverse. Then the lift to it exists by Proposition 1.2.1, which serves as a lift of the original map. The case of right n horn is similar. Conversely if $N(\mathcal{C})$ is a kan complex then for any morphism $f : x \to y$, and consider the lifting of $\Lambda^2_0 \xrightarrow{(f,1_x)} N(\mathcal{C})$ to Δ^2, which exists by assumption. This lift gives the left inverse of f. One can construct a right inverse by considering the right two horn corresponding to $(f, 1_y)$. Thus f is an isomorphism. $\square$

1.3 Examples

We begin with examples of ∞-categories. On an important note every ∞-category is a composer. However there are composers which are not ∞-categories. We will produce some such examples as well.

1.3.1 Nerve of a Category

Proposition 2.2.2 asserts that for an ordinary category category $\mathscr{C}$, the nerve $N(\mathscr{C})$ is an ∞-category. In particular Δ^n is an ∞-category as it is just $N([n])$.

1.3.2 ∞-category associated to a simplicial set

To any simplicial set we can associate an ∞-category. Given a simplicial set $\mathbf{X}$ consider the map $\mathbf{X} \to \Delta^0$. By small object argument applied to the set of inner horn inclusions one can factor this map into an inner fibration followed by an inner annodyne map i.e.$\exists$ a factorization $\mathbf{X} \to \mathcal{C}_{\mathbf{X}} \to \Delta^0$ of the above map. Since the later map is an inner fibration, we have that $\mathcal{C}_{\mathbf{X}}$ is an ∞-category. Moreover this gives a functor from the category of simplicial sets to its full sub category of ∞-categories since small object argument produces functorial factorization.

1.3.3 Composer associated to a simplicial set

Just similar to to what we did above for ∞-categories we can do the same by applying small object argument to spine inclusions and associate a composer to every simplicial set.

1.3.4 Singular homology of a topoogical space

Given a topological space $\mathbf{X}$ its singular homology is the simplicial set $S(\mathbf{X})$ whose n simplicies are given by

$$S(\mathbf{X})_n = Top(|\Delta^n|, X) \tag{1.5}$$

Where $|\Delta^n|$ is the standard topological n simplex. S extends to a functor $S : Top \to sSets$ which has a left adjoint namely the geometric realization. Clearly a horn lifting problem for $S(\mathbf{X})$ in $sSet$is nothing but a topological horn lifting problem for $\mathbf{X}$ in Top . But since the topological horn inclusions has retracts, such a lifting problem solves immediately. Therefore $S(\mathbf{X})$ thus defined is a Kan complex, in particular it is an ∞-category.

1.3.5 2-nerve of a 2 category

For $n \geq 0$ consider the 2-category $\mathbf{2}^n$ whose object set is given by $\{0, 1, 2, ..., n\}$ and the hom between two object i, j is given by the poset: $\mathbf{2}^n(i, j) = \{I \subset [0, n] \mid i, j \in I\}$ ordered by inclusion. Now given an arbitrary 2-category $\mathscr{C}$ the 2-nerve of it is given by $\mathcal{N}(\mathscr{C})_n = Fun(\mathbf{2}^n, \mathscr{C})$, where the right hand side is the set of strict 2-functors of 2-categories. Clearly by this extends to an adjoint pair, $\mathbf{2} \dashv \mathcal{N}$. With some elementary computations in 2-categories it is easy to show that the 2-nerve of a 2 category is an ∞-category. Explicitly an n-simplex of the nerve consists of a chain of $n+1$ objects $\{C_0, C_1, ..., C_n\}$, a 1-morphism $f_{ij} : C_i \to C_j$ for each $i < j$ and a 2-morphism $\alpha_{ijk} : f_{ik} \to f_{jk} \circ f_{ij}$ with an obvious associativity condition on α's given on 4 indices. The lifting of a 2-horn can be done by putting the composition of 1-morphisms at the remaining edge and by putting an α in the interior accordingly. Any other higher dimensional interiors are just identities so the 2 horn lifting is enough.

1.3.6 ∞-category of topological spaces: Top^∞

As we mentioned in the introduction that the first example of infinity category was hoped to be topological spaces, lets mention it here now. An n-simplice of Top^∞ is given by a chain $\{X_0, X_1, ..., X_n\}$ of topological spaces and a $j - i - 1$ homotopy from X_i to X_j such that their restriction to each face is a $j - i - 2$ homotopy.

1.3.7 Classifying space of a group (monoid) : BG (BM)

The classifying space $\mathbf{B}G$ of a group is a Kan complex and is an ∞-category therefore. More generally, given an associative monoid M, define the simplicial set $\mathbf{B}M$ whose n-simplicies form the set M^n, degeneracies are given by the corresponding projections and ith faces are given by inclusions with the identity element at the ith position. With some effort one can show that this gives an ∞-category (which is a kan complex iff M is a group).

Chapter 2

Further Examples and construction of ∞ -categories

In the last chapter we discussed some few examples but they are kind of trivial. In this chapter we continue the discussion and provide some more general way of constructing new ∞-category from known ones. Some of these constructions are very fundamental to the theory and we will repeatedly use these constructions throughout. We begin with limits and colimits.

2.1 Closedness for Limits and Colimits of the subcategory of ∞Categories $\subset$sSet

Proposition 2.1.1. *Let I be a set . And let $\{\mathcal{C}_i\}_{i\in I}$ be a collection of simplicial sets. Then the product $\prod_{i\in I}\mathcal{C}_i$, in the category sSet is an $\infty-$category iff each $\mathcal{C}_i$ is an $\infty-$category.*

Proof. Consider the lifting problem for $0 < j < n$

$$\begin{tikzcd} \Lambda^n_j \arrow[r, "g"] \arrow[d, hook, "x"'] & \prod_{i\in I} \mathcal{C}_i \\ \Delta^n \arrow[ur, dotted] & \end{tikzcd}$$

But composing with the universal maps defining the product we immediately have a lift for each $i \in \mathcal{I}$ since $\mathcal{C}_i$ is an ∞-category.

$$\begin{tikzcd} \Lambda^n_j \arrow[r] \arrow[d, hook] & \prod_{i\in I} \mathcal{C}_i \arrow[d, "p_i"] \\ \Delta^n \arrow[r, dotted, "a_i"] & \mathcal{C}_i \end{tikzcd}$$

Clearly the maps a_i thus constructed by universal property of product of product implies, $\exists! a : \Delta^n \to \prod_{i\in I} \mathcal{C}_i$ such that $p_i a = a_i$ for each i. But then one immediately has $p_i a x = a_i x = p_i g$ for each i , whence $ax = g$. Thus a itself factorizes the initial diagram providing a lift to the initial diagram.

Conversely consider the same lifting problem for some k as indicated in the left diagram below.

$$2 \quad \begin{tikzcd} \Lambda^n_j \arrow[r, "g"] \arrow[d, hook, "x"'] & \mathcal{C}_k \\ \Delta^n \arrow[ur, dotted] & \end{tikzcd} \qquad\qquad \begin{tikzcd} \Lambda^n_j \arrow[r, dashed, "g'"] \arrow[rr, bend left, "g"] \arrow[d, hook, "x"'] & \prod_{i\in I} \mathcal{C}_i \arrow[r, "p_k"] & \mathcal{C}_k \\ \Delta^n \arrow[ur, dotted] & & \end{tikzcd} \tag{2.1}$$

Choose some n simplex for each $\mathcal{C}_i$ $\forall i \neq k$, i.e. a map $v_i : \Delta^n \to \mathcal{C}_i$ which restricted to Λ^n_j gives a map $v_i x : \Lambda^n_j \to \mathcal{C}_i$. These maps along with g gives a unique map $g' : \Lambda^n_j \to \prod_{i\in I} \mathcal{C}_i$ such that the top part of the right diagram commutes. If $\prod_{i\in I} \mathcal{C}_i$ is an ∞-category then there is a diagram as in diagram on the right which serves as a lift to the original diagram. This proves that each $\mathcal{C}_k$ is an ∞-category if the product is. $\square$

Proposition 2.1.2. *Let $F : \omega \to sSet$ be a filtered diagram such that each $F(i)$ is an*

∞-category. Then $\varinjlim_{i\in\omega} F(i)$ is an ∞-category

Proof. Since Λ^n_j are small objects in the category of simplicial sets any map from Λ^n_j to a ω filtered colimit will actually factor through one of the objects and that object being an ∞-category will guarantee that object will have a lift which composed with the universal coproduct map gives a lift for the original diagram. $\square$

Proposition 2.1.3. *Let I be a set . And let $\{\mathcal{C}_i\}_{i\in I}$ be a collection of ∞-categories. Then the coproduct $\coprod_{i\in I}\mathcal{C}_i$ in the category $sSet$ is an $\infty-$category iff each component is an $\infty-$category.*

Proof. Suppose that each of the components are ∞-categories. Consider the lifting problem for $0 < j < n$

$$\begin{tikzcd}
\Lambda^n_j \arrow[r, "g"] \arrow[d, hook, "x"'] & \coprod_{i\in I}\mathcal{C}_i \\
\Delta^n \arrow[ur, dotted] &
\end{tikzcd}$$

Now Λ^n_j is connected for any j, the map g therefore must land in exactly one (say the kth component) component providing a lift for the original diagram as that object itself an ∞-category.

$$\begin{tikzcd}
\Lambda^n_j \arrow[r] \arrow[d, hook, "x"'] & \mathcal{C}_k \arrow[r, "q_i"] & \coprod_{i\in I}\mathcal{C}_i \\
\Delta^n \arrow[ur, dashed] \arrow[urr, dotted, "a"] & &
\end{tikzcd}$$

Conversely if $\coprod_{i\in I}\mathcal{C}_i$ is an ∞-category then look at a lifting problem for $\mathcal{C}_k$ for some $k \in I$. And consider the composition with the universal coproduct map. By assumption a lift indicated by dotted arrow in the above diagram exists. Once again since Δ^n is connected, a must land inside exactly one component But since the top component land inside $\mathcal{C}_k$ so will have to be the case for the bottom map. Which means the dashed lift also exists. $\square$

2.2 Coherent nerve of a simplicially enriched category

Theoretically Simplicially enriched categories ($sSet - Cat$) are as good model of higher categories as ∞-category. In practice these can be used interchangeable. In this section we will discuss one side of this correspondence namely given a simplicially enriched category we will associate an ∞-category to it in some universal way. This construction will be called coherent nerve. However we will start with its left adjoint which makes much more sense to define, via free resolution construction.

Let $\mathbf{X}$ be a simplicial set we want to associate a simplicially enriched category to it. A wild guess for this should be a category whose objects are just the set $\mathbf{X}_0$. And the simplicial hom between two object $x, y \in \mathbf{X}_0$ be given by:

$$Hom(x, y)_n = \{\sigma : \Delta^{n+1} \to \mathbf{X} \mid \sigma_{|\{0\}} = x, \sigma_{|\{n+1\}} = y\} \subseteq \mathbf{X}_n \tag{2.2}$$

However this does not quite satisfy us as it does not give all the paths from x to y. For example let $\mathbf{X} = \Delta^2$ and $x = 0, y = 2$ then there should be two element in $Hom(0, 2)$ namely by the straight edge and another via the vertex 1. Therefore we must add more elements to the above definition via concatenation of simplicies.

$$Hom(x, y)_n = \left\{\vee_{i=0}^{i=l}\sigma_i \mid \sigma : \Delta^{n_i} \to \mathbf{X}, \sigma_{0|\{0\}} = x, \sigma_{l|\{n_l\}} = y, \sigma_{i+1|\{0\}} = \sigma_{i|\{n_i\}}\right\}/\sim \tag{2.3}$$

Where the product is meant for concatenation. The equivalence relation must come from identification of these $n + 1$ simplicies when they agree on the n implicies. However the description seems to be quite complicated. Such processes in mathematics are usually called free resolution and has a nice functorial presentation. We therefore describe the functorial presentation of the above via simplicial free resolution. Consider the category $Graph^*$ of directed graphs with a special choice of self loop for each vertex. There is a obvious forgetful functor $U : Cat \to Graph^*$ from the category of small categories to $Graph^*$ which sends a category to the graph of morphisms and objects with identity on each object to the chosen special loop the graph. This functor has a left adjoint F which sends a graph to the corresponding free category where the marked edge becomes the identity. In this way we get a comonad $FU : Cat \to Cat$ which is identity on objects and thus a functor $C^! : Cat \to sSet - Cat$. The left kan extension of this along the usual nerve functor $N : Cat \to sSet$ exists and is called the Homotopy coherent nerve functor. Explicitly it is

given by the kan resolution:

$$\mathfrak{C}(\mathbf{X}) = \varinjlim_{\Delta^n \downarrow \mathbf{X}} C^!([n]) \tag{2.4}$$

One immediately observes that $\mathfrak{C} \circ N \cong C^!$. Moreover :

Proposition 2.2.1. *Let $\mathbf{X}$ be an ∞-category. For any two object $x, y \in \mathbf{X}_0$ the simplicial hom of the resolution $\mathfrak{C}(\mathbf{X})$ is a composer. (Note that it need not be an ∞-category in general)*

Proof. The proof is elementary. Just concatenate the consecutive n necklaces in $\mathfrak{C}(\mathbf{X})(x, y)_n$ to get a filler along Δ^n. □

It is clear that $\mathfrak{C}$ preserves colimits. It has some more properties as listed below. We refer to [8] for details of the proofs.

Lemma 2.2.2. *1. $\mathfrak{C}$ preserves products up to weak equivalence.*

2. $\mathfrak{C}$ takes monomorphisms to cofibrations in $sSet - Cat$

Since $\mathfrak{C}$ preserves colimits it has a right adjoint called the homotopy coherent nerve, and is given by:

$$\mathfrak{N}(\mathcal{C})_n = sSet - Cat(\mathfrak{C}(\Delta^n), \mathcal{C}) \tag{2.5}$$

Proposition 2.2.3. *Let $\mathcal{C}$ be a simplicially enriched category. Then the homotopy coherent nerve $\mathfrak{N}(\mathcal{C})$ is a composer.*

Proof. Consider the following lifting problems .

$$\begin{tikzcd} I^n \arrow[r] \arrow[d, hook] & \mathfrak{N}(\mathcal{C}) \\ \Delta^n \arrow[ur, dotted] & \end{tikzcd} \qquad \begin{tikzcd} \mathfrak{C}(I^n) \arrow[r] \arrow[d, hook] & \mathcal{C} \\ \mathfrak{C}(\Delta^n) \arrow[ur, dotted] & \end{tikzcd} \tag{2.6}$$

By nerve-resolution adjunction, the left lifting problem solves iff the right one solves. But to solve the right one it suffices to show that the map $\mathfrak{C}(I^n) \to \mathfrak{C}(\Delta^n)$ has a retract. We

claim that $\mathfrak{C}(I^n) \cong c([n])$,the constant simplicial category. Firstly,$\mathfrak{C}(I^1) = \mathfrak{C}(\Delta^1) = c([1])$. By assuming the claim to be true for $n = n$ we get,

$$\mathfrak{C}(I^{n+1}) = \mathfrak{C}(I^n \coprod_{\Delta^0} I^1) = \mathfrak{C}(I^n) \coprod_{\mathfrak{C}(\Delta^0)} \mathfrak{C}(I^1) = c(n) \coprod_{c(\Delta^0)} c(1) = c([n] \coprod_{[0]} [1]) = c([n+1]) \quad (2.7)$$

Thus its indeed true by induction. Now the obvious functor $\pi : \mathfrak{C}(\Delta^n) \to c[n]$ is identity on objects where as the map $c([n]) \cong \mathfrak{C}(I^n) \to \mathfrak{C}(\Delta^n)$ is also identity on objects. Thus composing the two maps we get a composite from $c([n])$ to itself which is identity on objects. But $c([n])$ is totally ordered whence any endofunctor is determined by its action on objects. Therefor the above composite is just identity on $c([n])$ and hence on $\mathfrak{C}(I^n)$. $\square$

Theorem 2.2.4. *Let $\mathcal{C}$ be a Kan enriched category. Then the homotopy coherent nerve $\mathfrak{N}(\mathcal{C})$ is an ∞-category.*

Proof. Let $0 < j < n$ and consider the following lifting problems:

$$\begin{tikzcd} \Lambda^n_j \arrow[r] \arrow[d, hook] & \mathfrak{N}(\mathcal{C}) \\ \Delta^n \arrow[ur, dotted] & \end{tikzcd} \qquad \begin{tikzcd} \mathfrak{C}(\Lambda^n_j) \arrow[r, "g"] \arrow[d, hook] & \mathcal{C} \\ \mathfrak{C}(\Delta^n) \arrow[ur, dotted] & \end{tikzcd} \quad (2.8)$$

By nerve-resolution adjunction, the left lifting problem solves iff the right one solves. Since on objects $\mathfrak{C}$ is just identity, the right lifting problem has to be solved only on simplicial homs. That is one is left with solving the following lifting problem for each $0 \leq i \leq k \leq n$,

$$\begin{tikzcd} \mathfrak{C}(\Lambda^n_j)(i,k) \arrow[r, "g"] \arrow[d, hook] & \mathcal{C}(g(i), g(k)) \\ \mathfrak{C}(\Delta^n)(i,k) \arrow[ur, dotted] & \end{tikzcd}$$

By lemma 2.2.2, $\mathfrak{C}$ takes monomorphism to cofibrations in $sSet - Cat$. Therefore the vertical map in the above is actually an inclusion. Moreover both the homs $\mathfrak{C}(\Lambda^n_j)(i,k)$ and $\mathfrak{C}(\Delta^n)(i,k)$ are contractible, so the map is also a weak equivalence.Indeed applying $\mathfrak{C}$ to the coequalizer diagram defining the horns one finds that the homs $\mathfrak{C}(\Lambda^n_j)(i,k)$ indicate a horn on a corner of the cube $\mathfrak{C}(\Delta^n)(i,k)$. Thus the left vertical map in the above diagrams, is

anodyne. If the category is Kan enriched then the right hand side hom is a kan complex, and hence the lifting problem solves by definition of anodyne maps. □

2.3 Exponentials

The idea of this section is to prove that the simplicial hom $\mathcal{C}^{\mathbf{X}}$ is an ∞-category for $\mathcal{C}$ any ∞-category and $\mathbf{X}$ any simplicial set. We begin with the following construction. Given maps of simplicial sets $A \xrightarrow{i} B$ and $S \xrightarrow{g} T$ and $X \xrightarrow{f} Y$ we have the following commutative diagrams.

$$\begin{tikzcd} A \times S \arrow[r, "i\times 1"] \arrow[d, "1\times g"'] & B \times S \arrow[d, "1\times g"] \\ A \times T \arrow[r, "i\times 1"] & B \times T \end{tikzcd} \qquad \begin{tikzcd} X^B \arrow[r] \arrow[d] & X^A \arrow[d] \\ Y^B \arrow[r] & Y^A \end{tikzcd} \tag{2.9}$$

And therefore $\exists!$ maps,

$$i \otimes g : A \times T \coprod_{A\times S} B \times S \to B \times T \tag{2.10}$$

$$\langle f, i\rangle : X^B \to X^A \times_{Y^A} Y^B \tag{2.11}$$

Lemma 2.3.1. *The two lifting problems mentioned below are equivalent.*

$$\begin{tikzcd} A \times T \coprod_{A\times S} B \times S \arrow[r] \arrow[d, "i\otimes g"'] & X \arrow[d, "f"] \\ B \times T \arrow[r] \arrow[ur, dotted] & Y \end{tikzcd} \qquad \begin{tikzcd} S \arrow[r] \arrow[d, "g"'] & X^B \arrow[d, "\langle f,i\rangle"] \\ T \arrow[r] \arrow[ur, dotted] & X^A \times_{Y^A} Y^B \end{tikzcd} \tag{2.12}$$

Proof. This follows from the adjunction $U \times _ \dashv (_)^U$, which is natural in U. □

Lemma 2.3.2. *Let i and g be monomorphisms. Then $i \otimes g$ is a monomorphism. Moreover $i \otimes g$ is (right, left, inner) anodyne if either i or g is (right, left, inner) anodyne.*

Proof. We will show the case of inner anodyne maps. Without loss of generality assume that i is an inner anodyne map . The class of maps g , such that $i \otimes g$ is inner anodyne, i.e. has left lifting property w.r.t inner fibrations (IFB) has left lifting property w.r.t. $\langle IFB, i\rangle$.

Therefore such g's form a class= $llp\langle IFB, i\rangle$ and hence is a saturated class. Thus to prove the lemma we need to show that $i \otimes (\partial\Delta^n \hookrightarrow \Delta^n)$ is inner anodyne $\forall n$, provided i is inner anodyne. But the class of maps i for which $i \otimes (\partial\Delta^n \hookrightarrow \Delta^n)$ is inner anodyne is again saturated. Thus we are left to show that $(\Lambda^m_j \hookrightarrow \Delta^m) \otimes (\partial\Delta^n \hookrightarrow \Delta^n)$ is inner anodyne for any $n, m, 0 < j < m$ which is a standard result in simplicial homotopy theory. □

Proposition 2.3.3. *Let $i : A \to B$ be a monomorphism and $f : X \to Y$ a (inner, left, right) fibration. Then,*

1. *$\langle f, i\rangle$ is a (inner, left, right) fibration*

2. *Moreover $\langle f, i\rangle$ is a trivial fibration if i is (inner, left, right)anodyne.*

Proof. To prove 1 for the case of inner fibration, we need to show that $\langle f, i\rangle$ has right lifting property with respect to inner anodyne maps. Let then g be any inner anodyne map. We need to to show that any diagram on the right as below has a lift.

$$\begin{tikzcd} A \times T \coprod_{A\times S} B \times S \arrow[r] \arrow[d, "i\otimes g"] & X \arrow[d, "f"] \\ B \times T \arrow[r] \arrow[ur, dotted] & Y \end{tikzcd} \qquad \begin{tikzcd} S \arrow[r] \arrow[d, "g"] & X^B \arrow[d, "\langle f,i\rangle"] \\ T \arrow[r] \arrow[ur, dotted] & X^A \times_{Y^A} Y^B \end{tikzcd} \tag{2.13}$$

But by lemma 3.3.1 the right diagram has a lift iff the left one has. But if g is inner anodyne, we have just shown that so is $i \otimes g$. So the left diagram has a lift since f is an inner fibration. To prove 2, we need to solve the same lifting problem when g is any monomorphism and i is an inner anodyne map. Again by the previous lemma $i \otimes g$ is inner anodyne. Then the right lifting problem solves since f is an inner fibration. □

Theorem 2.3.4. *Let $\mathcal{C}$ be an ∞-category (Kan complex respectively), and $i : A \to B$ be a monomorphism. Then the induced map $\mathcal{C}^B \to \mathcal{C}^A$ is an inner fibration (Kan fibration respectively). In particular, $\mathcal{C}^B$ is an ∞-category (Kan complex respectively), for any simplicial set B.*

Proof. The proof of the first statement follows from the first part of the previous lemma by putting $Y = \Delta^0$. Whereas the last claim follows from the first claim by putting $A = \phi$. □

2.4 Join and Slice

In this section we discuss the idea of formally adding an ∞-category to anther ∞-category either to the domain or to the codomain. The idea is similar to the join construction mentioned at the next subsection. We would also like to talk about the idea of a 'slice category' for ∞-categories for whatever it means and would prove that the resultants are indeed ∞-categories. The idea of slice is a replacement of the notion of a slice category in ordinary category as described below.

2.4.1 For 1-categories

Given two ordinary 1-categories $\mathscr{C}$ and $\mathscr{D}$ one can form the join $\mathscr{C} * \mathscr{D}$ by taking as objects $ob(\mathscr{C}) \coprod ob(\mathscr{D})$ and attaching a unique map from each object of $\mathscr{C}$ to each object of $\mathscr{D}$. rest of the structures are kept as they are. Thus $\mathscr{C} * \mathscr{D}$ is the gluing of $\mathscr{D}$ to $\mathscr{C}$ in the codomain or $\mathscr{C}$ to $\mathscr{D}$ in the domain. There are obvious inclusion maps $\mathscr{C} \hookrightarrow \mathscr{C} * \mathscr{D} \hookleftarrow \mathscr{D}$. However this is clearly not symmetric thing. Therefore we get a functor: $Cat \xrightarrow{\mathscr{C}*-} Cat_{\mathscr{C}/}$ which has a right adjoint given by the slice functor, which sends a functor F below $\mathscr{C}$ to the category of $\mathrm{CCone}(F)$ cocones on the functor whose objects are cocones on F and morphisms are morphisms from cocone object to cocone object which are compatible with the cocone maps. If $\mathrm{Codomain}(F) = \mathscr{E}$, then we write, $\mathrm{CCone}(F) = \mathscr{E}_{F/}$ and call it the over slice category of F. Similar to this if one considers the other functor namely $Cat \xrightarrow{-*\mathscr{D}} Cat_{\mathscr{D}/}$ the right adjoint is given by cones on a functor with domain $\mathscr{D}$ and is denoted $\mathscr{E}_{/F} = \mathrm{Cone}(F)$, called the above slice category. Note that since colimits are universal cocone and limits are universal cones, the above construction is very fundamental in the theory of ordinary categories. As expected the same will be the case for ∞-categories as well.

2.4.2 For ∞-categories

In this section we give a general description of join and slice for any pair of arbitrary simplicial sets, which will serve the purpose for ∞-categories as well.

Definition 2.4.1. *Given two simplicial sets X and Y we define the simplicial join $X * Y$*

as follows: its set of n simplices is given by-

$$(X * Y)_n = \coprod_{i=0}^{i=n+1} X_{i-1} \times Y_{n-i} \tag{2.14}$$

where we define $X_{-1} = Y_{-1} = \Delta^0$. In other words,

$$(X * Y)_n = X_n \coprod (X_0 \times Y_{n-1}) ..... \coprod (X_{n-1} \times Y_0) \coprod Y_n \tag{2.15}$$

Equation 3.15 immediately gives inclusion maps $X \hookrightarrow X*Y \hookleftarrow Y$ since $X_n, Y_n \subset (X*Y)_n$. This description defines a functors, $_ * X, X * _ : sSet \to sSet_{X/}$, the fact that these functor preserves colimits are elementary computations. By adjoint functor theorem therefore they both have right adjoint simply by kan resolution. On objects $p : X \to Y$ the right adjoint of $X * _$ is denoted $Y_{p/}$ and the right adjoint of $_ * X$ is denoted $Y_{/p}$ and are given by,

$$(Y_{p/})_n = sSet_{X/}(X * \Delta^n, Y) \tag{2.16}$$

$$(Y_{/p})_n = sSet_{X/}(\Delta^n * X, Y) \tag{2.17}$$

Lemma 2.4.1. *Every join $X * Y$ comes with a canonical map to Δ^1. Conversely any map of simplicial sets $p : K \to \Delta^1$, must factor through one such canonical map.*

Proof. According to definition 3.14, every element x of $(X * Y)_n$ corresponds to some $0 \leq i \leq n + 1$. Just send x to the map $[n] \to [1]$ which takes $j \leq i$ to 0 and rest to 1. On the other hand, the obvious decomposition of p is is just via the inverse images $p^{-1}(0) =: K_0$ and $p^{-1}(1) =: K_1$. In fact, this decomposition is universal and functorial over Δ^1. □

Theorem 2.4.2. *Let X and Y be simplicial sets. Then $X * Y$ is an ∞-category iff both X and Y are ∞-categories.*

Proof. Assume that $X * Y$ is an ∞-category. We prove that X is an ∞-category. Consider a lifting problem for X (the left diagram) ,

$$\begin{tikzcd} \Lambda^n_j \arrow[r, "g"] \arrow[d, hook, "i"] & X \\ \Delta^n \arrow[ur, dotted] & \end{tikzcd} \qquad \begin{tikzcd} \Lambda^n_j \arrow[r, "g"] \arrow[d, hook, "i"] & X \arrow[r, "i_X"] & X * Y \arrow[d, "p"] \\ \Delta^n \arrow[ur, dotted] \arrow[urr, dashed, "a"] & & \Delta^1 \end{tikzcd} \tag{2.18}$$

Since $X * Y$ is an ∞-category the dashed lift exists on the right. Consider the canonical map p to Δ^1 for $X * Y$. Clearly $pi_X g$ is constant at 0 (as it factors through i_X). But then pa is constant at 0. This implies that a factors through i_X, i.e. the dashed lift exists (as i_X is a monomorphism). Proving the case of Y is similar, one uses i_Y instead of i_X. Conversely assume that both X and Y are ∞-categories and consider a lifting problem as in the left diagram below.

$$\begin{tikzcd} \Lambda^n_j \arrow[r, "g"] \arrow[d, hook] & X * Y \\ \Delta^n \arrow[ur, dotted] & \end{tikzcd} \qquad \begin{tikzcd} & & (\Lambda^n_j)_0 * (\Lambda^n_j)_1 \arrow[dl, dashed, "a"] \arrow[ddl, bend left] \\ \Lambda^n_j \arrow[r, "g"] \arrow[d, hook] \arrow[urr, bend left] & X * Y \arrow[d, "p"] & \\ \Delta^n \arrow[r, dashed, "x"] & \Delta^1 & \end{tikzcd} \tag{2.19}$$

Let p be the canonical map $X * Y$ to Δ^1 as in lemma 3.14. Again by lemma 3.14 the factorization of pg via is given by $(\Lambda^n_j)_0 * (\Lambda^n_j)_1$. However since Δ^1 is the nerve of a category the dashed lift in the right diagram exists. Clearly any such map x is just a canonical map $\Delta^k * \Delta^{n-k-1} \to \Delta^1$ where $k \in \{-1, 0, ..., n\}$ where the first $k+1$ entry map to 0 and the rest maps to 1 and we let $\Delta^{-1} = \phi$, If $k = -1$ then x is constant at 1 and hence so is $xi = pg$, in which case $(\Lambda^n_j)_0 = \phi$ and $(\Lambda^n_j)_1 = \Delta^n$ or if $k = n$ then $(\Lambda^n_j)_1 = \phi$ and $(\Lambda^n_j)_0 = \Delta^n$. So we have that if $k = -1, n$, then $(\Lambda^n_j)_0 * (\Lambda^n_j)_1 = (\phi * \Delta^n \ or \ \Delta^n * \phi) = \Delta^n$ then the m simplicies of $(\Lambda^n_j)_0$ are all those m simplcies of Δ^n which has image in $\{0, ..., k\}$ and is an m simplex of Λ^n_j. But since $k \neq n$ the last condition is redundant. Whence the m simplicies of $(\Lambda^n_j)_0$ are precisely those m simplcies of Δ^n which has image in $\{0, ..., k\}$ which is Δ^k. Thus $(\Lambda^n_j)_0 = \Delta^k$ similarly $(\Lambda^n_j)_1 = \Delta^{n-k-1}$ i.e. $(\Lambda^n_j)_0 * (\Lambda^n_j)_1 = \Delta^k * \Delta^{n-k-1} = \Delta^n$ if $k \neq -1, n$.Thus altogether $\forall k, (\Lambda^n_j)_0 * (\Lambda^n_j)_1 = \Delta^n$. And a serves as a lift to the original diagram. $\square$

We will now talk about the concept of slice as defined by the right adjoint of join. The functoriality of the slice as an operator is still not clear. We will do that in the following way. Consider $Tw(sSet)$ the twisted arrow category of simplicial sets: the objects are simplicial set maps $A \to B$ and a morphism (f, i) from $B \xrightarrow{\phi} X$ to $A \xrightarrow{f'} Y$ is a commutative diagram:

$$\begin{tikzcd} A \arrow[r, "f'"] \arrow[d, "i"] & Y \\ B \arrow[r, "\phi"] & X \arrow[u, "f"'] \end{tikzcd}$$

Define $\Psi : Tw(sSet) \to sSet$, the slice functor as follows:

$$\begin{aligned}
&\text{on objects : } \Psi(B \xrightarrow{\phi} X) = X_{\phi/} \\
&\text{on morphisms } (f,i)\text{as above :}\Psi(f,i) : X_{\phi/} \to Y_{f'/} = Y_{f\phi i/}
\end{aligned}$$

which is adjoint to the map, $A * X_{\phi/} \xrightarrow{i * X_{\phi/}} B * X_{\phi/} \to X \xrightarrow{f} Y$, where the middle map is the counit of the corresponding adjunction. The functoriality of Ψ is obvious. Now we mention the notion of join pushout product mimicking the idea of usual pushout product, discussed in the previous section. Given maps of simplicial sets $A \xrightarrow{i} B \xrightarrow{\phi} X \xrightarrow{f} Y$ and $S \xrightarrow{g} T$ one gets the following commutative diagrams:

$$\begin{tikzcd}
A * S \arrow[r, "i*1"] \arrow[d, "1*g"'] & B * S \arrow[d, "1*g"] \\
A * T \arrow[r, "i*1"] & B * T
\end{tikzcd}
\qquad
\begin{tikzcd}
X_{\varphi/} \arrow[r] \arrow[d] & X_{\varphi i/} \arrow[d] \\
Y_{f\varphi/} \arrow[r] & Y_{f\varphi i/}
\end{tikzcd} \tag{2.20}$$

Once again by universal property of pull back and push out we get the following unique maps:

$$i \circledast g : A * T \coprod_{A*S} B * S \to B * T \tag{2.21}$$

$$[f,i] : X_{\varphi/} \to X_{\varphi i/} \times_{Y_{f\varphi i/}} Y_{f\varphi/} \tag{2.22}$$

Lemma 2.4.3. *The two lifting problems mentioned below are equivalent.*

$$\begin{tikzcd}
B \arrow[r] \arrow[rr, bend left, "\phi"] & A * T \coprod_{A*S} B * S \arrow[r] \arrow[d, "i\circledast g"'] & X \arrow[d, "f"] \\
& B * T \arrow[r] \arrow[ur, dotted] & Y
\end{tikzcd}
\qquad
\begin{tikzcd}
S \arrow[r] \arrow[d, "g"'] & X_{\varphi/} \arrow[d, "{[f,i]}"] \\
T \arrow[r] \arrow[ur, dotted] & X_{\varphi i/} \times_{Y_{f\phi i/}} Y_{f\phi/}
\end{tikzcd} \tag{2.23}$$

Proof. The proof is again via join slice adjunction and universal properties of pullbacks and pushouts. □

Lemma 2.4.4. *Let i and g be monomorphisms then $i \circledast g$ is a monomorphism. Moreover:*

1. *if i is right anodyne or g is left anodyne,then $i \circledast g$ is inner anodyne*

2. *if f is left anodyne,then $i \circledast g$ is left anodyne*

3. *if g is right anodyne, then $i \circledast g$ is right anodyne.*

In other words for a monomorphism i, $i \circledast _$ preserves right anodyne map but takes left anodyne maps to inner anodyne maps. Similarly given monomorphism g, $_ \circledast g$ preserves left anodyne maps but takes right anodyne maps to inner anodyne maps.In particular both $i \circledast _$, $_ \circledast g$ preserve inner anodyne map.

Proof. We will prove only one case here as well. Let f be a right anodyne map and g any monomorphism. To show that $i \circledast g$ is inner anodyne it is enough to show that for any $n, m, 0 < j \leq n$ $(\Lambda^n_j \hookrightarrow \Delta^n) \circledast (\partial\Delta^m \hookrightarrow \Delta^m)$ is an inner anodyne map (This follows from the same method used in lemma 2.3.2). That is the join pushout map, $\Lambda^n_j * \Delta^m \coprod_{\Lambda^n_j * \partial\Delta^m} \Delta^n * \partial\Delta^m \longrightarrow \Delta^n * \Delta^m$ is inner anodyne. Using the fact that joins preserve colimits and that both boundary and horns are defined using certain coequalizer of coproducts of standard simplexes, one computes that the above map is just the horn inclusion $\Lambda^{n+M+1}_j \hookrightarrow \Delta^{n+M+1}$ and since $j \leq n < n+1 \leq n+m+1$, this is in fact an inner horn and thus inner anodyne. □

Proposition 2.4.5. *Let $A \xrightarrow{i} B \xrightarrow{\varphi} X \xrightarrow{f} Y$ be a given a sequence of maps, where i is a monomorphism. If f is a (inner, right)-fibration then the slice map $[f, i]$ is (left, right)-fibration. Moreover $[f, i]$ is a trivial fibration if either f is a trivial fibration or i is inner anodyne. The dual results for the other slice also holds.*

Proof. We show only one case the other cases are similar. Let then f be an inner fibration. We need to show that the lifting problem on the right solves whenever g is inner anodyne.

$$\begin{tikzcd}
B \arrow[r] \arrow[rrr, bend left, "\phi"] & A * T \coprod_{A*S} B * S \arrow[rr] \arrow[d, "i \circledast g"'] & & X \arrow[d, "f"] \\
& B * T \arrow[rr] \arrow[urr, dotted] & & Y
\end{tikzcd}
\qquad
\begin{tikzcd}
S \arrow[r] \arrow[d, "g"'] & X_{\phi/} \arrow[d, "{[f,i]}"] \\
T \arrow[r] \arrow[ur, dotted] & X_{\phi i/} \times_{Y_{f\phi i/}} Y_{f\phi/}
\end{tikzcd} \tag{2.24}$$

But by lemma 3., the lifting problem on the right solves iff the lifting problem on the left solves. Now by lemma 3.1 the map $i \circledast g$ is inner anodyne as g is inner anodyne by assumption. Therefore the lifting problem on the left solves since f is an inner fibration. □

Theorem 2.4.6. *For any sequence $A \xrightarrow{i} B \xrightarrow{\phi} \mathcal{C}$ where i is a monomorphism of simplicial sets and $\mathcal{C}$ is an ∞-category the induced map $\mathcal{C}_{\varphi/} \to \mathcal{C}_{\varphi i/}$ is a left fibration and $\mathcal{C}_{/\varphi} \to \mathcal{C}_{/\varphi i}$ is a right fibration. In particular for any map $\phi : B \to \mathcal{C}$ the canonical maps $\mathcal{C}_{\varphi/} \to \mathcal{C}$ and $\mathcal{C}_{/\varphi} \to \mathcal{C}$ are left and right fibrations respectively. Moreover both $\mathcal{C}_{/\varphi}$ and $\mathcal{C}_{\varphi/}$ are ∞-categories.*

Proof. For the first part, put $X = \mathcal{C}$,$Y = \Delta^0$ and f as the unique map to Δ^0, in the previous proposition. The second part is just a special case of the first part for $A = \phi$. To solve the last part Consider the lifting problem on the left for some inner horn inclusion.

$$\begin{tikzcd} \Lambda^n_j \arrow[r, "g"] \arrow[d, hook, "i"] & \mathcal{C}_{\varphi/} \\ \Delta^n \arrow[ur, dotted] & \end{tikzcd} \qquad \begin{tikzcd} \Lambda^n_j \arrow[r, "g"] \arrow[d, hook, "i"] & \mathcal{C}_{\varphi/} \arrow[d] \\ \Delta^n \arrow[r, dashed, "a"] \arrow[ur, dotted] & \mathcal{C} \end{tikzcd} \tag{2.25}$$

Since $\mathcal{C}$ is an ∞-category the extension a as in the right diagram exists. But by the second part we know that the right most vertical line is a left fibration and thus an inner fibration, whence the right diagram admits a lift, which immediately serves as a lift to to the original diagram, implying that $\mathcal{C}_{\varphi/}$ is an ∞-category. The case of the other slice is similar. $\square$

2.5 Some practical examples of ∞ categories

2.5.1 ∞-category of abstract spaces: Spc

Consider the category Kan of kan complexes with the simplicial homs given by usual simplicial home. Clearly this category is kan enriched as simplicial homs of kan complexes are kan complexes. The homotopy coherent nerve of this category $\mathfrak{N}(Kan)$ is an $\infty - category$ which is called the ∞-category of spaces and is denoted Spc.

2.5.2 ∞-category of concrete spaces: $\widehat{Spc}$

Let CW denote the category of CW complexes as objects and simplicial homs being defined by the singular homology of usual mapping homs of continuous maps given open compact

topology. The nerve of this category is also called the ∞-category of spaces and is denoted $\widehat{Spc}$. The last two categories are Joyal equivalent so we can use them interchangeably.

2.5.3 Categories enriched over groupoids

Every ordinary 1-category is a kan enriched by viewing each hom as a constant simplicial set. Its Homotopy coherent nerve is the same as its usual nerve when viewed as an ordinary 1-category. In general for a category $\mathscr{C}$ enriched over groupoids (for example the category of presheaves or sheaves of groupoids on a site), one can apply the usual nerve functor in the homs and get a category enriched over kan complexes (proposition 1.2.5). The coherent nerve of this category is then an ∞-category .

2.5.4 Homotopy ∞ -category of an abelian category

Given an abelian category $\mathscr{A}$ consider the category of bounded below chain complexes $\mathrm{Ch}_{\geq 0}(\mathscr{A})$ in $\mathscr{A}$. Clearly this category is enriched over the the category of bounded below chain complexes of abelian groups, where the hom between two maps $C_\bullet \underset{g}{\overset{f}{\rightrightarrows}} D_\bullet$ is given by the chain complex on mapping space induced by a chain homotopy. Let $\mathcal{K}$ be the following composite:

$$Ch_{\geq 0}(\mathbb{Z}) \xrightarrow{\Gamma} Fun(\Delta^{op}, Ab) \hookrightarrow sSet$$

where the first map Γ is the Dold-Kan equivalence between chain complexes of abelian groups and simplicial abelian groups. And the second map is the obvious forgetful one. Since the category $Ch_{\geq 0}(\mathscr{A})$ is a category enriched over $Ch_{\geq 0}(\mathbb{Z})$, by applying $\mathcal{K}$ to the homs we get a simplicially enriched category $\mathcal{K}(Ch_{\geq 0}(\mathscr{A}))$ which is clearly kan enriched as every simplicial abelian group is a kan complex. Therefore the coherent nerve of this category is an ∞-category . We call it the ∞-homotopy category of $\mathscr{A}$. Clearly it is a refinement of the ordinary homotopy category to the language of ∞-categories.

Chapter 3

Morphisms and equivalences

In this chapter we will talk about various structures in ∞-categories which arise naturally in the theory. The beauty of these would lie in the fact that just like in the ordinary category of Sets every construction gives a set, every construction will lead to an ∞-category , if nor we will take groupoid cores to make it an ∞-category or will take the ∞-categorical replacement as in example 2.3.2.

3.1 composing morphisms

As we have mentioned earlier there is a simplicial set of ways to compose two morphisms in a composer. We emphasise that such a simplicial set becomes interesting in the case of an ∞-category.

Since $I^2 = \Lambda^2_1 = \Delta^1 \coprod_{\Delta^0} \Delta^1$ where we identify the domain of the second and source of the first, given a simplicial set $\mathbf{X}$ we have $\mathbf{X}^{\Lambda^2_1} = \mathbf{X}^{\Delta^1} \prod_{\mathbf{X}} \mathbf{X}^{\Delta^1}$. But then a map from the the inner two horn to $\mathbf{X}$ is just a 0 simplice in $\mathbf{X}^{\Delta^1} \prod_{\mathbf{X}} \mathbf{X}^{\Delta^1}$, which is just a pair of 1 simplices $f, g \in \mathbf{X}_1$ such that $d_1(g) = d_0(f)$. Now consider the inner 2 horn inclusion $\Lambda^2_1 \hookrightarrow \Delta^2$. Which immediately gives rise to the map $\mathbf{X}^{\Delta^2} \to \mathbf{X}^{\Lambda^2_1}$.

Definition 3.1.1. *Given two morphisms f, g in $\mathbf{X}_1$ such that $d_1(g) = d_0(f)$ the composition*

of f and g is defined to be the pullback of the following diagram :

$$\begin{tikzcd} comp_{\mathbf{X}}(f,g) \arrow[r] \arrow[d] & \mathbf{X}^{\Delta^2} \arrow[d] \\ \Delta^0 \arrow[r, "(f,g)"] & \mathbf{X}^{\Lambda^2_1} \end{tikzcd} \tag{3.1}$$

where the map (f,g) is the the 0 simplex in $\mathbf{X}^{\Lambda^2_1} = \mathbf{X}^{\Delta^1} \prod_{\mathbf{X}} \mathbf{X}^{\Delta^1}$ given by the unique map arising by the definition of pull back:

$$\begin{tikzcd} \Delta^0 \arrow[dr, dotted, "(f,g)"] \arrow[drr, bend left, "f"] \arrow[ddr, bend right, "g"'] & & \\ & \mathbf{X}^{\Delta^1} \prod_{\mathbf{X}} \mathbf{X}^{\Delta^1} \arrow[r] \arrow[d] & \mathbf{X}^{\Delta^1} \arrow[d, "d_0"] \\ & \mathbf{X}^{\Delta^1} \arrow[r, "d_1"] & \mathbf{X} \end{tikzcd}$$

Clearly the 0 simplices of $comp_{\mathbf{X}}(f,g)$ are precisely the lifts of horns in $\mathbf{X}$ along Δ^2.

Proposition 3.1.1. *If $\mathcal{C}$ is an ∞-category then the simplicial set $comp_{\mathcal{C}}(f,g)$ is also an ∞-category and in fact it is a contractible kan complex. Therefore in an ∞-category there is a unique composition up to a contractible kan complex.*

Proof. Since $\Lambda^2_1 \hookrightarrow \Delta^2$ is inner anodyne, by proposition 2.3.3 $\mathcal{C}^{\Delta^2} \to \mathcal{C}^{\Lambda^2_1}$ is trivial fibration. But trivial fibrations are stable under pull back, thus the left most map in diagram 4.1 defining composition is a trivial fibration whence $comp_{\mathbf{X}}(f,g)$ becomes a contractible kan complex. □

In a composer one can define a notion of $n-$composition in general for n many morphisms by similar process, by considering the n spine inclusion instead of the 2-spine inclusion. Note that since the spine inclusions are inner anodyne and every ∞-category is a composer the n composition for an ∞-category is again going to be a contractible kan complex. We conclude this section by depicting a proof for this claim.

3.2 Equivalences in ∞-Categories

Once it is known how to compose morphisms the next question one should ask is whether such a composition admits inverses. The answer is easy: if σ is a 2 simplex which restricts to f on $\{0,1\}$, g on $\{1,2\}$ and 1 on $\{0,2\}$ then 1 is a composition of g and f and we can call g a left inverse of f and f a right inverse of g.

Definition 3.2.1. *A 1- simplex f in a simplicial set $\mathbf{X}$ is called and equivalence if it admits both a left and a right inverse. Or, equivalently if both the lifting problem:*

$$\begin{tikzcd} \Lambda^2_0 \arrow[r, "(f,1)"] \arrow[d, hook] & \mathbf{X} \\ \Delta^2 \arrow[ur, dotted] & \end{tikzcd} \qquad \begin{tikzcd} \Lambda^2_2 \arrow[r, "(1,f)"] \arrow[d, hook] & \mathbf{X} \\ \Delta^2 \arrow[ur, dotted] & \end{tikzcd} \tag{3.2}$$

solves. Or, equivalently the image of f in the homotopy category of $\mathbf{X}$ is an isomorphism i.e. f is equivalent to identity both on source and target.

Definition 3.2.2. *An ∞-category is called an ∞-groupoid if every morphism in it is an equivalence or equivalently the homotopy category is a groupoid in the classical sense.*

Given an ordinary category $\mathscr{C}$, by $\mathscr{C}^{\simeq}$ we denote the subcategory of isomorphisms. Sine the notion of equivalences in an ∞-category is based on the homotopy category we can easily use the following pullback diagram to construct an ∞-category which will contain all the equivalences in a given ∞-category $\mathcal{C}$.

$$\begin{tikzcd} \mathcal{C}^{\simeq} \arrow[r] \arrow[d] & \mathcal{C} \arrow[d] \\ N((h(\mathcal{C}))^{\simeq}) \arrow[r] & Nh(\mathcal{C}) \end{tikzcd}$$

Note that on 1 simplices $\mathcal{C}^{\simeq}$ consists of exactly those morphisms of $\mathcal{C}$ which are equivalences, and we all it the **maximal sub** $-\infty$ **groupoid** or the groupoid core of $\mathcal{C}$.

Proposition 3.2.1. *If $\mathcal{C}$ is an ∞-category then the maximal sub∞-groupoid is an ∞-category and is, in fact, an ∞-groupoid.*

Given a lifting problem for $\mathcal{C}^{\simeq}$ one deduces lifts for $N((h(\mathcal{C}))^{\simeq})$ and $\mathcal{C}$ and the outer

diagrams commute since there can be a single lift for $Nh(\mathcal{C})$ as it is a nerve.Then by universal property of pullback a lift to $\mathcal{C}^{\simeq}$ is deduced.

$$\begin{tikzcd}
\Lambda^n_j \arrow[r] \arrow[d, hook] & \mathcal{C}^{\simeq} \arrow[r] \arrow[d] & \mathcal{C} \arrow[d] \\
\Delta^n \arrow[ur, dotted] \arrow[r, dashed] & N((h(\mathcal{C}))^{\simeq}) \arrow[r] & Nh(\mathcal{C})
\end{tikzcd}$$

Now we show that the homotopy category of $\mathcal{C}$ is a groupoid. But morphisms of $\mathcal{C}^{\simeq}$ are all the equivalances of $\mathcal{C}$, and hence $h(\mathcal{C}^{\simeq}) = (h\mathcal{C})^{\simeq}$ which is already a groupoid by definition.

Definition 3.2.3. *An inner fibration $\mathcal{C} \xrightarrow{p} \mathcal{D}$ of ∞-categories is called conservative if $p(f)$ is an equivalence in $\mathcal{D}$ implies that f is an equivalence in $\mathcal{C}$.*

Definition 3.2.4. *An inner fibration $\mathcal{C} \xrightarrow{p} \mathcal{D}$ of ∞-categories is called an isofibration if the left or equivalently the right diagram where the bottom map indicates an equivalence of $\mathcal{D}$ has a lift which is an equivalence of $\mathcal{C}$.*

$$\begin{tikzcd}
\{0\} \arrow[r] \arrow[d] & \mathcal{C} \arrow[d, "p"] \\
\Delta^1 \arrow[r] \arrow[ur, dotted] & \mathcal{D}
\end{tikzcd}
\qquad\qquad
\begin{tikzcd}
\{1\} \arrow[r] \arrow[d] & \mathcal{C} \arrow[d, "p"] \\
\Delta^1 \arrow[r] \arrow[ur, dotted] & \mathcal{D}
\end{tikzcd} \tag{3.3}$$

Proposition 3.2.2. *Right or left fibrations are conservative isofibrations.*

Proof. We show the case for left fibrations. Let then $\mathcal{C} \xrightarrow{p} \mathcal{D}$ be a left fibration of ∞-categories and let f be a morphism in $\mathcal{C}$ such that $p(f)$ is an equivalence. Consider the lifting problem in $\mathcal{C}$:

$$\begin{tikzcd}
\Lambda^2_0 \arrow[r, "(f,1)"] \arrow[d, hook] & \mathcal{C} \arrow[d, "p"] \\
\Delta^2 \arrow[r, dashed] \arrow[ur, dotted] & \mathcal{D}
\end{tikzcd}$$

Since $p(f)$ is an equivalence in $\mathcal{D}$ the lifting problem solves for $\mathcal{D}$. But p is a left fibration. So we get a lift for the whole diagram. Which gives a right inverse g of f. Applying the same argument to g itself we see that g has a right inverse as well. Thus g is an equivalence in $\mathcal{C}$ and hence f is an equivalence in $\mathcal{C}$. Now a lift to the left diagram in definition 3.2.4 always exists, as $\{0\} \to \Delta^1$ is the left horn inclusion. Since we have already shown that p is conservative we are done. $\square$

Theorem 3.2.3. ***Joyal's lifting theorem.***
Let $\mathcal{C} \xrightarrow{p} \mathcal{D}$ be an inner fibration of ∞-categories. The following lifting problems, where ϕ is an equivalence in $\mathcal{C}$ has solutions.

$$\begin{tikzcd} \Delta^1 \arrow[r, "\{0,1\}"'] \arrow[rr, bend left, "\phi"] & \Lambda^n_0 \arrow[r] \arrow[d, hook] & \mathcal{C} \arrow[d, "p"] \\ & \Delta^n \arrow[r] \arrow[ur, dotted] & \mathcal{D} \end{tikzcd} \qquad \begin{tikzcd} \Delta^1 \arrow[r, "\{n-1,n\}"'] \arrow[rr, bend left, "\phi"] & \Lambda^n_n \arrow[r] \arrow[d, hook] & \mathcal{C} \arrow[d, "p"] \\ & \Delta^n \arrow[r] \arrow[ur, dotted] & \mathcal{D} \end{tikzcd} \tag{3.4}$$

Proof. We solve the case for the right diagram.
We note that $\Lambda^n_n \hookrightarrow \Delta^n$ is isomorphic to the join push out map:
$\Lambda^n_n = \partial\Delta^{n-2} * \Delta^1 \cup \Delta^{n-2} * \Lambda^1_1 = \partial\Delta^{n-2} * \Delta^1 \coprod_{\partial\Delta^{n-2}*\Lambda^1_1} \Delta^{n-2} * \Lambda^1_1 \longrightarrow \Delta^{n-2} * \Delta^1 = \Delta^n$ (this follows from applying the colimit preserving join functor to the coequalizer diagrams defining Λ^{n-2}_0 and $\partial\Delta^1$). Then by lemma 2.4.3 this lifting problem has a solution iff the following lifting problem solve (i is the inclusion $\partial\Delta^{n-2} \hookrightarrow \Delta^{n-2}$)

$$\begin{tikzcd} \{1\} \arrow[r] \arrow[d] & \mathcal{C}_{\phi/} \arrow[d, "{[p,i]}"] \\ \Delta^1 \arrow[r, "\psi"] \arrow[ur, dotted] & \mathcal{C}_{\phi(1)/} \times_{\mathcal{D}_{p\phi(1)/}} \mathcal{D}_{p\phi/} \end{tikzcd} \tag{3.5}$$

Now consider the following sequence of maps where the middle square is a pull back :

$$\begin{tikzcd} \mathcal{C}_{\phi/} \arrow[r] & \mathcal{C}_{\phi(1)/} \times_{\mathcal{D}_{p\phi(1)/}} \mathcal{D}_{p\phi/} \arrow[r, "v"] \arrow[d] & \mathcal{C}_{\phi(1)/} \arrow[r, "u"] \arrow[d] & \mathcal{C} \\ & \mathcal{D}_{p\phi(1)/} \arrow[r] & \mathcal{D}_{p\phi(1)/} & \end{tikzcd}$$

Due to proposition 2.4.5 and theorem 2.4.6 the first and third top horizontal maps are left fibrations. But the bottom horizontal map is also a left fibration due to the same and therefore so is the top middle horizontal map being the pull back of the lower one.Thus the top composite uv is a left fibration. Now consider the following extension of the diagram 4.5

above.

$$\begin{tikzcd}
\{1\} \arrow[r] \arrow[d] & \mathcal{C}_{\phi/} \arrow[d, "{[p,i]}"] \\
\Delta^1 \arrow[r, "\psi"] \arrow[ur, dotted] \arrow[dr, "\phi"'] & \mathcal{C}_{\phi(1)/} \times_{\mathcal{D}_{p\phi(1)/}} \mathcal{D}_{p\phi/} \arrow[d, "uv"] \\
& \mathcal{C}
\end{tikzcd}$$

Now $uv(\psi) = \phi$ is an equivalence in $\mathcal{C}$ which implies that ψ is an equivalence as uv is a left fibration and thus conservative. Coming back to diagram 4.5, we see that $[p, i]$ is an isofibration (since it is a left fibration) and ψ is an equivalence. Thus the lift in diagram 4.5 do exists by definition of isofibration, and the proof is complete. $\square$

Theorem 3.2.4. *Let p be an inner fibration $\mathcal{C} \xrightarrow{p} \mathcal{D}$ of ∞-categories . Then p is conservative iff any lifting problem on the left where $p(\phi)$ is an equivalence in $\mathcal{D}$ has a solution. Dually, p is conservative iff any lifting problem on the right where $p(\phi)$ is an equivalence in $\mathcal{D}$ has a solution.*

$$\begin{tikzcd}
\Delta^1 \arrow[r, "\{0,1\}"'] \arrow[rr, bend left, "\phi"] & \Lambda^n_0 \arrow[r] \arrow[d, hook] & \mathcal{C} \arrow[d, "p"] \\
& \Delta^n \arrow[r] \arrow[ur, dotted] & \mathcal{D}
\end{tikzcd} \qquad \begin{tikzcd}
\Delta^1 \arrow[r, "\{n-1,n\}"'] \arrow[rr, bend left, "\phi"] & \Lambda^n_n \arrow[r] \arrow[d, hook] & \mathcal{C} \arrow[d, "p"] \\
& \Delta^n \arrow[r] \arrow[ur, dotted] & \mathcal{D}
\end{tikzcd} \tag{3.6}$$

Proof. If p is conservative then $p(\phi)$ is an equivalence would imply that ϕ is an equivalence, whence by Joyal's lifting theorem the required lifts exist. Conversely consider the case where left diagrams have lifts for each n . Let ϕ be a morphism in $\mathcal{C}$ such that $p(\phi)$ is an equivalence in $\mathcal{D}$. We want to lift the diagram on the left below, to find a right inverse. But $p(\phi)$ is an equivalence, yielding a lift for the right diagram by the dashed line.

$$\begin{tikzcd}
\Lambda^2_0 \arrow[r, "(\phi,1)"] \arrow[d, hook] & \mathcal{C} \\
\Delta^2 \arrow[ur, dotted] &
\end{tikzcd} \qquad \begin{tikzcd}
\Lambda^2_0 \arrow[r, "(\phi,1)"] \arrow[d, hook] & \mathcal{C} \arrow[d, "p"] \\
\Delta^2 \arrow[r, dashed] \arrow[ur, dotted] & \mathcal{D}
\end{tikzcd} \tag{3.7}$$

By assumption, then the dotted lift also exists, providing a lift for the original diagram providing a right inverse ψ for ϕ.Applying the same idea to ψ we get a right inverse for ψ.

Thus ψ becomes an equivalence and therefore so is ϕ. $\square$

Corollary 1. *An ∞-category is a kan complex iff it is an ∞-groupoid.*

Proof. According to definition, to show that $\mathcal{C}$ is an ∞-groupoid, one needs to show that the two lifting problems in diagram 4.2 solves for $\mathbf{X} = \mathcal{C}$. But this is immediate if $\mathcal{C}$ is a kan complex. Conversely if $\mathcal{C}$ is an ∞-groupoid, then we need to find a lift to the following diagram,

$$\begin{tikzcd} \Lambda^n_0 \arrow[r] \arrow[d, hook] & \mathcal{C} \\ \Delta^2 \arrow[ur, dotted] & \end{tikzcd} \qquad \begin{tikzcd} \Lambda^n_n \arrow[r] \arrow[d, hook] & \mathcal{C} \\ \Delta^2 \arrow[ur, dotted] & \end{tikzcd} \tag{3.8}$$

Applying theorem 3.2.4 to the conservative (as domain is an ∞-groupoid) inner fibration $\mathcal{C} \xrightarrow{p} \Delta^0$, we get the desired lifts. $\square$

One may deduce an immediate one line proof of proposition 1.2.5 from the above corollary: $N(\mathcal{C})$ is a kan complex iff $N(\mathcal{C})$ is an ∞-groupoid iff $h(N(\mathcal{C})) = \mathcal{C}$ is a groupoid!

Theorem 3.2.5. *Any inner fibration $K \xrightarrow{f} \mathcal{D}$ where k is a kan complex, is a kan fibration.*

Apply theorem 3.2.4. The conditions on morphisms are satisfied immediately. Hence lift to any horn inclusion exists by theorem 3.2.4.

Theorem 3.2.6. *A inner fibration of ∞-categories is an isofibration iff it is a kan fibration when restricted to the the groupoid cores.*

Proof. Let $\mathcal{C} \xrightarrow{p} \mathcal{D}$ be an isofibration of ∞-categories. Referring to the previous theorem it suffices to show that $\mathcal{C}^{\simeq} \xrightarrow{p^{\simeq}} \mathcal{D}^{\simeq}$ is an isofibration. For arbitrary n and $0 < j < n$ consider the lifting problem on the left square of the diagram below.

$$\begin{tikzcd} \Lambda^n_j \arrow[r] \arrow[d, hook] & \mathcal{C}^{\simeq} \arrow[r, hook] \arrow[d] & \mathcal{C} \arrow[d, "p"] \\ \Delta^n \arrow[r] \arrow[ur, dotted] \arrow[urr, dashed, "a"] & \mathcal{D}^{\simeq} \arrow[r, hook] & \mathcal{D} \end{tikzcd}$$

Since $\mathcal{C} \xrightarrow{p} \mathcal{D}$ is an inner fibration, the dashed arrow exists. Now the n-simplex a of $\mathcal{C}$ has all of its edges in $\mathcal{C}^{\simeq}$, as edges are already in the horn. But then this implies that a actually

lands in $\mathcal{C}$. It remains to prove that $p^{\simeq}$ is an isofibration. But this follows from the fact a lift for p exists which being an equivalence also belongs to $\mathcal{C}^{\simeq}$. The dotted lift is actually the dashed one.

$$\begin{tikzcd}
\{0\} \arrow[r] \arrow[d] & \mathcal{C}^{\simeq} \arrow[r, hook] \arrow[d] & \mathcal{C} \arrow[d, "p"] \\
\Delta^1 \arrow[r] \arrow[ur, dotted] \arrow[urr, dashed, "a"] & \mathcal{D}^{\simeq} \arrow[r, hook] & \mathcal{D}
\end{tikzcd}$$

Conversely if $p^{\simeq}$ is a kan fibration of the groupoid cores consider the lifting where g is an equivalence in $\mathcal{D}$. Since the 0-simplices of $\mathcal{C}$ and $\mathcal{C}^{\simeq}$ are the same to say that g is an equivalence in $\mathcal{D}$ is to say that the given diagram factors through $p^{\simeq}$ and we get a lift (the dotted lift) for the left square, as $p^{\simeq}$ is a kan fibration giving a lift (the dashed lift) for the whole diagram.

$$\begin{tikzcd}
\{0\} \arrow[r] \arrow[d] & \mathcal{C} \arrow[d] \\
\Delta^1 \arrow[r, "g"'] & \mathcal{D}
\end{tikzcd}
\qquad\qquad
\begin{tikzcd}
\{0\} \arrow[r] \arrow[d] & \mathcal{C}^{\simeq} \arrow[r, hook] \arrow[d] & \mathcal{C} \arrow[d, "p"] \\
\Delta^1 \arrow[r, "g"'] \arrow[ur, dotted] \arrow[urr, dashed] & \mathcal{D}^{\simeq} \arrow[r, hook] & \mathcal{D}
\end{tikzcd} \tag{3.9}$$

□

Proposition 3.2.7. *If $\mathcal{C} \xrightarrow{p} \mathcal{D}$ is a trivial fibration of ∞-categories then the restriction to the groupoid cores $\mathcal{C}^{\simeq} \xrightarrow{p^{\simeq}} \mathcal{D}^{\simeq}$ is a trivial fibration as well. This holds even if $\mathcal{C}$ and $\mathcal{D}$ are any simplicial set.*

Proof. Consider the lifting problem for the left square below:

$$\begin{tikzcd}
\partial\Delta^n \arrow[r] \arrow[d, hook] & \mathcal{C}^{\simeq} \arrow[r, hook] \arrow[d] & \mathcal{C} \arrow[d, "p"] \\
\Delta^n \arrow[r] \arrow[ur, dotted] \arrow[urr, dashed, "x"] & \mathcal{D}^{\simeq} \arrow[r, hook] & \mathcal{D}
\end{tikzcd}$$

Since the whole diagram is commutative and p is a trivial fibration the dashed lift x exists always. If $n \geq 2$ then $I^n \subset \partial\Delta^n$ whence the spine of x is in $\mathcal{C}^{\simeq}$ and therefore x itself is in $\mathcal{C}^{\simeq}$ and the lifting problem solves. If $n = 1$, then note that p being a trivial fibration is a left fibration in particular and thus is conservative. But then $x : \Delta^1 \to \mathcal{C}$ is a morphism in $\mathcal{C}$ such that $px \in \mathcal{D}^{\simeq}$ or an equivalence in $\mathcal{D}$. Which implies that $x \in \mathcal{C}$, since p is conservative, and the lifting problem solves. □

Proposition 3.2.8. *Let $\mathcal{C}$ be an ∞-category. The map $\mathcal{C}^{\Delta^1} \xrightarrow{i^*} \mathcal{C} \times \mathcal{C}$ induced by the boundary inclusion $\partial\Delta^1 \hookrightarrow \Delta^1$ is conservative.*

Proof. Due to theorem 2.3.4 we know that i^* is an inner fibration. Now let $\Delta^1 \xrightarrow{f} \mathcal{C}^{\Delta^1}$ be a one simplex such that $i^*(f)$ is an equivalence in $\mathcal{C} \times \mathcal{C}$. We need to show that f is an equivalence in $\mathcal{C}^{\Delta^1}$. For that consider the following diagram:

$$\begin{tikzcd}
\Lambda^2_0 \arrow[r, "(f,1)"] \arrow[d, hook] & \mathcal{C}^{\Delta^1} \\
\Delta^2 \arrow[ur, dotted] &
\end{tikzcd}$$

Since $i^*(f)$ is an equivalence the dashed line exists in the following diagram. We claim that the dotted lift also exists in the below diagram, which serves as a lift for the original diagram.

$$\begin{tikzcd}
\Lambda^2_0 \arrow[r, "(f,1)"] \arrow[d, hook] & \mathcal{C}^{\Delta^1} \arrow[d, "i^*"] \\
\Delta^2 \arrow[r, dashed] \arrow[ur, dotted] & \mathcal{C} \times \mathcal{C}
\end{tikzcd}$$

By lemma 2.3.1 this is the case iff the following lift exists.

$$\begin{tikzcd}
\partial\Delta^1 \times \Delta^2 \cup \Delta^1 \times \Lambda^2_0 \arrow[r] \arrow[d, hook] & \mathcal{C} \arrow[d, "i"] \\
\Delta^1 \times \Delta^2 \arrow[r] \arrow[ur, dotted] & \{0\}
\end{tikzcd}$$

$$\begin{tikzcd}
 & & 5 \\
3 \arrow[r] \arrow[urr] & 4 \arrow[ur, dashed] & \\
 & & 2 \arrow[uu] \\
0 \arrow[uu] \arrow[r] \arrow[urr] \arrow[uur, dotted] & 1 \arrow[uu] \arrow[ur, dashed] \arrow[uur, dotted] &
\end{tikzcd}$$

Now consider the simplex as above, which is exactly the simplex $\Delta^1 \times \Delta^2$ with $\{0,1\}$ and $\{3,4\}$ are equivalences in $\mathcal{C}$. The left up corner is given by all the faces except $\{1,2,4,5\}$. We will lift this diagram step by step. First of all, the simplex {0,2,5}, {1,4,5}, {0,1,5} has fillers as inner 2 horns lift. Now in the simplex {0,1,2,5} con-

sider the zero horn. Since the first edge {0,1} is an equivalence in $\mathcal{C}$ the filler exists due to Joyal's lifting theorem.A filler for the simplex {0,1,4,5} also exists due to the same reason. The simplex {0,3,4,5} has a filler as at the vertex 3 it is given as an inner horn. Thus we have filled up the whole simplex in $\mathcal{C}$. □

Proposition 3.2.9. *Let K be an arbitrary simplicial set and let $sk_0(K)$ be the 0-skeleton. Then for an ∞-category $\mathcal{C}$ the functor $\mathcal{C}^K \xrightarrow{i^*} \mathcal{C}^{sk_0(K)} = \prod_{x \in K_0} \mathcal{C}$ is conservative.*

Proof. We know that $K = \varinjlim_{n<\omega} sk_n(K)$. Thus $\mathcal{C}^K = \varprojlim_{n<\omega} \mathcal{C}^{sk_n(K)}$. Therefore referring to theorem 4.2.4 i^* is conservative if each map $\mathcal{C}^{sk_n(K)} \xrightarrow{i_n^*} \mathcal{C}^{sk_0(K)}$ is conservative. For $n \geq 1$ the following diagram on the left is a push out diagram. The right one is therefore a pullback.

$$\begin{tikzcd} \coprod \partial\Delta^n \arrow[r] \arrow[d] & sk_{n-1}(K) \arrow[d] \\ \coprod \Delta^n \arrow[r] & sk_n(K) \end{tikzcd} \qquad \begin{tikzcd} \mathcal{C}^{sk_n(K)} \arrow[r] \arrow[d] & \prod \mathcal{C}^{\partial\Delta^n} \arrow[d] \\ \mathcal{C}^{sk_{n-1}(K)} \arrow[r] & \mathcal{C}^{\Delta^n} \end{tikzcd} \tag{3.10}$$

Since conservative fibrations are stable under countable limits, the left vertical maps would be conservative if if each map $\mathcal{C}^{\partial\Delta^n} \to \partial\Delta^n$ are. Let's consider the case for $n \geq 2$. In this situation the spine inclusion $I^n \to \Delta^n$ factors through the boundary inclusion and hence we have a sequence $\mathcal{C}^{\Delta^n} \to \mathcal{C}^{\partial\Delta^n} \to \mathcal{C}^{I^n}$ which composes to a conservative map as the spine inclusion is inner anodyne and so the composite above is a trivial fibration.This implies that the first map of this sequence is conservative. Finally the case of $n = 1$ is just the previous proposition. □

Theorem 3.2.10. *Let $K \xrightarrow{f} L$ be a morphism of simplicial sets which induce bijection on 0-simplices, then $\mathcal{C}^L \xrightarrow{f^*} \mathcal{C}^K$ is conservative.*

Proof. In the following commutative diagram, both the vertical maps and the bottom map are conservative, this implies that the top one is also conservative.

$$\begin{tikzcd} \mathcal{C}^L \arrow[r, "f^*"] \arrow[d] & \mathcal{C}^K \arrow[d] \\ \mathcal{C}^{L_0} \arrow[r] & \mathcal{C}^{K_0} \end{tikzcd} \tag{3.11}$$

□

Theorem 3.2.11. *Let $K \xrightarrow{f} L$ be a morphism of simplicial sets which induce bijection on 0-simplices, then for any inner fibration $\mathcal{C} \xrightarrow{p} \mathcal{D}$ the induced map on power objects, $\mathcal{C}^L \xrightarrow{\langle p,f\rangle} \mathcal{C}^K \times_{\mathcal{D}^K} \mathcal{D}^L$ is conservative.*

Proof. Since the composite $\mathcal{C}^L \xrightarrow{\langle p,f\rangle} \mathcal{C}^K \times_{\mathcal{D}^K} \mathcal{D}^L \longrightarrow \mathcal{C}^K$ is conservative due to theorem 4.2.9, the functor $\langle p, f\rangle$ reflects equivalences. Since $\langle p, f\rangle$ is already an inner fibration it must be conservative. □

Theorem 3.2.12. *Let $K \xrightarrow{f} L$ be a monomorphism of simplicial sets which induce bijection on 0-simplices, then for any inner fibration $\mathcal{C} \xrightarrow{p} \mathcal{D}$ the induced map on power objects, $\mathcal{C}^L \xrightarrow{\langle p,f\rangle} \mathcal{C}^K \times_{\mathcal{D}^K} \mathcal{D}^L$ is an isofibration.*

Proof. The following diagram, where the bottom map is an equivalence, must have a lift for the claim to be true.

$$\begin{tikzcd}
\{0\} \arrow[r] \arrow[d] & \mathcal{C}^L \arrow[d, "\langle p,f\rangle"] \\
\Delta^1 \arrow[r] & \mathcal{C}^K \times_{\mathcal{D}^K} \mathcal{D}^L
\end{tikzcd}$$

But this lifting problem is equivalent to the below lifting problem due to lemma 2.3.3.

$$\begin{tikzcd}
K \times \Delta^1 \coprod_{K\times\{0\}} L \times \{0\} \arrow[r] \arrow[d, "f\otimes\{0\}"] & \mathcal{C} \arrow[d, "p"] \\
L \times \Delta^1 \arrow[r] \arrow[ur, dotted] & \mathcal{D}
\end{tikzcd}$$

Clearly the set of maps as in the theorem for which the above diagram solves is the left lifting class of the map $\langle p, \{0\}\rangle$ and is thus saturated. Therefore it is enough to show the case for the boundary inclusions $\partial\Delta^n \hookrightarrow \Delta^n$ for $n \geq 1$. A similar reasoning as in proposition 4.2.7 via a proper filtering does the job. □

Theorem 3.2.13. *Let $K \xrightarrow{f} L$ be a monomorphism of simplicial sets then for any isofibation $\mathcal{C} \xrightarrow{p} \mathcal{D}$ the induced map on power objects, $\mathcal{C}^L \xrightarrow{\langle p,f\rangle} \mathcal{C}^K \times_{\mathcal{D}^K} \mathcal{D}^L$ is an isofibration.*

Proof. Referring to the proof of the previous theorem it remains to show the case for the boundary inclusion for $n = 0$. That is the following lifting problem needs to be solved where

the bottom map is an equivalence,

$$\begin{tikzcd} \{0\} \arrow[r] \arrow[d] & \mathcal{C} \arrow[d, "p"] \\ \Delta^1 \arrow[r] \arrow[ur, dotted] & \mathcal{D} \end{tikzcd}$$

, which is immediate by the definition of p as an isofibration. $\square$

3.3 ∞-hom

Looking at the beginning of the chapter, the first question that one should ask is whether the morphisms of an ∞-category form an ∞-category. If not, is there a natural candidate for that? We answer this in the following section. Given a simplicial set $\mathbf{X}$ consider the maps $\mathbf{X}^{\Delta^1} \underset{d_0}{\overset{d_1}{\rightrightarrows}} \mathbf{X}^{\Delta^0} = \mathbf{X}$ given by domain and codomain of morphisms which gives a map $\mathbf{X}^{\Delta^1} \xrightarrow{(d_1,d_0)} \mathbf{X} \times \mathbf{X}$.

Definition 3.3.1. *Given a simplicial set $\mathbf{X}$ and two 0-simplex x and y the the simplicial mapping space betweeen x and y in $\mathbf{X}$ is defined to be the following pullback:*

$$\begin{tikzcd} hom^{sim}_{\mathbf{X}}(x,y) \arrow[r] \arrow[d] & \mathbf{X}^{\Delta^1} \arrow[d, "(d_1,d_0)"] \\ \Delta^0 \arrow[r, "(x,y)"] & \mathbf{X} \times \mathbf{X} \end{tikzcd}$$

Proposition 3.3.1. *If $\mathcal{C}$ is an ∞-category then for any two object x and y the simplicial mapping space $hom^{sim}_{\mathcal{C}}(x,y)$ is a kan complex. We denote this hom as $\mathcal{C}^{\infty}_1(x,y)$.*

Proof. The right most map in the above diagram is induced by the inclusion $\partial\Delta^1 \hookrightarrow \Delta^1$ thus by prop 4.2.7 it is conservative. Since conservative maps are stable under pullback we find that so is the left one. Since Δ^0 is an ∞-groupoid, by definition of conservative map the simplicial set $\mathcal{C}^{\infty}_1(x,y)$ is a groupoid and hence a kan complex. $\square$

The notation already suggests that one can define $\mathcal{C}^{\infty}_j$ homs in general. This can be done by taking pullbacks along the map $\mathcal{C}^{\Delta^n} \xrightarrow{(d_n,d_0)} \mathcal{C} \times \mathcal{C}$. Moreover, this homes can be defined for any number of end points. Not only that one can safely define composition in the

following way: given 3 objects x, y, z in a simplicial set $\mathbf{X}$ one has the following sequence:

$$hom^{sim}_{\mathbf{X}}(x,y) \times hom^{sim}_{\mathbf{X}}(y,z) \longrightarrow \mathbf{X}^{\Delta^1} \times_X \mathbf{X}^{\Delta^1} \longrightarrow \mathbf{X}^{\Delta^1}$$

composing to (x, z) after applying d_0, d_1. Thus by definition of $hom^{sim}_{\mathbf{X}}(x,z)$ as a pullback there is a unique map $hom^{sim}_{\mathbf{X}}(x,y) \times hom^{sim}_{\mathbf{X}}(y,z) \to hom^{sim}_{\mathbf{X}}(x,z)$ which we call the law of composition. In case of infinity category we say it is the ∞-law of composition. In particular given a morphism $y \xrightarrow{f} z$ this induce a map

$$hom^{sim}_{\mathbf{X}}(x,y) = hom^{sim}_{\mathbf{X}}(x,y) \times \Delta^0 \xrightarrow{1\times f} hom^{sim}_{\mathbf{X}}(x,y) \times hom^{sim}_{\mathbf{X}}(y,z) \longrightarrow hom^{sim}_{\mathbf{X}}(x,z)$$

and similarly for a map $x \xrightarrow{g} w$ a map

$$hom^{sim}_{\mathbf{X}}(w,y) = \Delta^0 \times hom^{sim}_{\mathbf{X}}(x,y) \xrightarrow{g\times 1} hom^{sim}_{\mathbf{X}}(x,w) \times hom^{sim}_{\mathbf{X}}(w,y) \longrightarrow hom^{sim}_{\mathbf{X}}(x,y)$$

Lemma 3.3.2. *If $y \xrightarrow{f} z$ is an equivalence in an ∞-category $\mathcal{C}$ then for any object x the induced maps: $\mathcal{C}^\infty_1(x,y) \to \mathcal{C}^\infty_1(x,z)$ and $\mathcal{C}^\infty_1(z,x) \to \mathcal{C}^\infty_1(y,x)$ are homotopy equivalences of kan complexes.*

Proof. There exists $z \xrightarrow{g} y$ such that $gf \sim 1$. The map $\mathcal{C}^\infty_1(x,z) \to \mathcal{C}^\infty_1(x,y)$ induced by g serves as the homotopy inverse to the map induced by f in the statement.

□

Definition 3.3.2. *An ∞-functor of ∞-categories is just a simplicial set map between them.*

Given an ∞-functor $\mathcal{C} \xrightarrow{f} \mathcal{D}$ and objects x, y in $\mathcal{C}$ the following diagram commutes:

$$\begin{tikzcd}
\mathcal{C}^\infty_1(x,y) \arrow[r] \arrow[d] & \mathcal{C}^{\Delta^1} \arrow[r] \arrow[d, "(d_1,d_0)"] & \mathcal{D}^{\Delta^1} \arrow[d, "(d_1,d_0)"] \\
\Delta^0 \arrow[r, "(x,y)"] & \mathcal{C}\times\mathcal{C} \arrow[r, "f\times f"'] & \mathcal{D}\times\mathcal{D}
\end{tikzcd}$$

the right hand one by inspection and the left one by definition. Thus by definition of $\mathcal{D}^\infty_1(fx, fy)$ there is a unique map $\mathcal{C}^\infty_1(x,y) \xrightarrow{f_{x,y}} \mathcal{D}^\infty_1(fx, fy)$ induced by f. The following definition is obvious.

Definition 3.3.3. *An ∞-functor $\mathcal{C} \xrightarrow{f} \mathcal{D}$ is called fully faithful if for any two object x, y the induced map $\mathcal{C}_1^\infty(x,y) \xrightarrow{f_{x,y}} \mathcal{D}_1^\infty(fx, fy)$ on infinity homs is a homotopy equivalence of kan complexes.*

Definition 3.3.4. *A full ∞-subcategory or a 1-subcategory $\mathcal{C}$, of an ∞-category $\mathcal{D}$ is a monomorphism of simplicial sets $\mathcal{C} \xrightarrow{f} \mathcal{D}$ such that f is fully faithful. An n-subcategory is such that for any two object x, y the induced map $\mathcal{C}_1^\infty(x,y) \xrightarrow{f_{x,y}} \mathcal{D}_1^\infty(fx, fy)$ is a $(n-1)$-subcategory.*

Definition 3.3.5. *Given a set S of objects in an ∞-category the full subcategory generated by S denoted S^∞ is the subsimplicial set whose 0-simplices set is exactly equal to S whereas the the set of 1-simplices are all morphism of the original ∞-category , whose domain and codomain lie in S. The higher simplices are only degenerate ones on the previous set.*

Definition 3.3.6. *let $f, g : \mathcal{C} \to \mathcal{D}$ be two ∞-functor .An ∞natural transformation η from an f to g is just a homotopy from f to g. Or equivalently a morphism $\eta : \Delta^1 \to \mathcal{D}^{\mathcal{C}}$ such that $\eta_{|\{0\}} = f$ and $\eta_{|\{1\}} = g$. An ∞natural transformation is called an ∞natural equivalence if it an equivalence as an edge in $\mathcal{D}^{\mathcal{C}}$.*

Theorem 3.3.3. *Let η be an ∞natural transformation from f to g of ∞-functors from an ∞-category $\mathcal{C}$ to an ∞-category $\mathcal{D}$. For any two object in $\mathcal{C}$ the following diagram with canonical maps, commutes up to homotopy. Moreover if η is a natural equivalence then the right vertical and bottom maps are homotopy equivalences*

$$\begin{tikzcd}
\mathcal{C}_1^\infty(x,y) \arrow[r, "f_{x,y}"] \arrow[d, "g_{x,y}"] & \mathcal{D}_1^\infty(fx, fy) \arrow[d] \\
\mathcal{D}_1^\infty(gx, gy) \arrow[r] & \mathcal{D}_1^\infty(fx, gy)
\end{tikzcd}$$

where the bottom map and the right vertical maps are given by η_{gy} and η_{fx} by pre and post composition respectively, which are homotopy equivalences if η is an ∞natural equivalence. In particular two naturally equivalent ∞-functor are homotopy equivalent on ∞homs.

Proof. This a straightforward computation, once noted that η is also given by a map $\mathcal{C} \xrightarrow{\eta} \mathcal{D}^{\Delta^1}$. The remaining part follows from lemma 4.3.2. $\square$

3.4 Equivalence of ∞-categories

Just like we have the collection of all categories forming another category Cat, same thing can be done in ∞-categories as well.
Let ∞Cat be the simplicially enriched category of all ∞-categories with homs given by the usual simplicial hom. Note that this category has no reason to be kan enriched so far and so the coherent nerve may not be an ∞-category. What we do is instead of the usual simplicial hom we take the groupoid cores of the simplicial homs. Since groupoid cores are kan complexes the category formed this way is now kan enriched.

Definition 3.4.1. *Let $\infty Cat^{\simeq}$ be the kan enriched category whose objects are ∞-categories and simplicial homs between two ∞-categories $\mathcal{C}$ to $\mathcal{D}$ is given by the groupoid cores $(\mathcal{D}^{\mathcal{C}})^{\simeq}$. The homotopy coherent nerve of this category denoted $Cat_\infty := \mathfrak{N}(\infty Cat^{\simeq})$, which is an ∞-category is called the infinity category of all infinity categories.*

Definition 3.4.2. ***Joyal Equivalence :*** *An ∞-functor is called a Joyal equivalence if its image in Cat_∞ is an equivalence (in the sense of equivalence in an ∞-category).*

Explicitly an ∞-functor $\mathcal{C} \xrightarrow{f} \mathcal{D}$ between ∞-categories is a Joyal equivalence if there is another ∞-functor $\mathcal{D} \xrightarrow{g} \mathcal{C}$ such that fg and gf are both homotopic to identities $1_{\mathcal{D}}$ and $1_{\mathcal{C}}$ where the homotopies are themselves equivalences in the ∞-categories $\mathcal{D}^{\mathcal{D}}$ and $\mathcal{C}^{\mathcal{C}}$ respectively. Clearly a Joyal equivalence between kan complexes is just a homotopy equivalence. The following corollary is immediate.

Corollary 2. *An ∞-functor $\mathcal{C} \xrightarrow{f} \mathcal{D}$ is fully faithful if for any two object x, y the induced map $\mathcal{C}_1^\infty(x,y) \xrightarrow{f_{x,y}} \mathcal{D}_1^\infty(fx, fy)$ on infinity homs is a Joyal equivalence.*

Theorem 3.4.1. *An ∞-functor $\mathcal{C} \xrightarrow{f} \mathcal{D}$ is a Joyal equivalence iff for any arbitrary ∞-category $\mathcal{E}$ the induced map $\mathcal{C}^{\mathcal{E}} \xrightarrow{f_*} \mathcal{D}^{\mathcal{E}}$ is a Joyal equivalence iff the induced map $\mathcal{E}^{\mathcal{D}} \xrightarrow{f^*} \mathcal{E}^{\mathcal{C}}$ is a Joyal equivalence of ∞-categories.*

Proof. Let an ∞-functor $\mathcal{C} \xrightarrow{f} \mathcal{D}$ be a Joyal equivalence of ∞-categories. Then there exists another ∞-functor $\mathcal{D} \xrightarrow{g} \mathcal{C}$ and homotopies $\alpha : \Delta^1 \to \mathcal{C}^{\mathcal{C}}$ from gf to $1_{\mathcal{C}}$ and $\beta : \Delta^1 \to \mathcal{D}^{\mathcal{D}}$ from fg to $1_{\mathcal{D}}$ such that they are themselves equivalences in the corresponding categories. It is straightforward to check that $(f^*, g^*, \alpha^*, \beta^*)$ and $(f_*, g_*, \alpha_*, \beta_*)$ assemble themselves into a similar kind of data for the corresponding categories. This implies that both f^* and f_* are

Joyal equivalences if f is.
Conversely assume that $\mathcal{C}^{\mathcal{E}} \xrightarrow{f_*} \mathcal{D}^{\mathcal{E}}$ is a Joyal equivalence for any arbitrary ∞-category $\mathcal{E}$.Consider the case of $\mathcal{E} = \mathcal{D}$. Since Joyal equivalence is stable under taking groupoid cores we have that $(\mathcal{C}^{\mathcal{D}})^{\simeq} \xrightarrow{f^*} (\mathcal{D}^{\mathcal{D}})^{\simeq}$ is also a Joyal equivalence which is just a homotopy equivalence. On π_0 this means that there exists 0-simplice in $(\mathcal{C}^{\mathcal{D}})^{\simeq}$ or a map $\mathcal{D} \xrightarrow{g} \mathcal{C}$ such that $f_*(g) \simeq 1_{\mathcal{D}}$ or, $fg \simeq 1_D$ where the homotopy itself is a 1-simplice in $(\mathcal{D}^{\mathcal{D}})^{\simeq}$. The same argument applied to $\mathcal{E} = \mathcal{C}$ results in $f_*(gf) = fgf \simeq f = f_*(1_{\mathcal{C}})$ which by injectivity of the homotopy equivalence f_* on π_0 implies that $gf \simeq 1_{\mathcal{C}}$ where once again the corresponding homotopy is a 1-simplice in $(\mathcal{C}^{\mathcal{C}})^{\simeq}$. Hence f is a Joyal equivalence. The f^* case follows similarly. □

Definition 3.4.3. *A map of simplicial sets $X \xrightarrow{f} Y$ is a Joyal equivalence if for any ∞-category $\mathcal{E}$ the induced map $\mathcal{E}^X \xrightarrow{f^*} \mathcal{E}^Y$ is a Joyal equivalence of ∞-categories.*

Note that due to theorem 3.4.1, the above definition does not do any harm to our previous definition concerning ∞-categories.

Proposition 3.4.2. *A trivial fibration of simplicial sets $X \xrightarrow{p} Y$ is a Joyal equivalence.*

Proof. Since every trivial fibration has a section, there exists $Y \xrightarrow{s} X$ such that $ps = 1_Y$. Clearly for any ∞-category $\mathcal{E}$ the induced map $(ps)^* : \mathcal{E}^Y \to \mathcal{E}^Y$ is equal to $1_{\mathcal{E}^Y}$.

On the other hand the map $X^X \xrightarrow{p_*} Y^X$ is also a trivial fibration. Due to proposition 4.2.7 $(X^X)^{\simeq} \xrightarrow{p_*^{\simeq}} (Y^X)^{\simeq}$ is also a trivial fibration and hence a bijection on π_0, in particular. But the objects sp and 1_X in are such that $p_*(sp) = psp = p = p_*(1_X)$. We must have $sp \simeq 1_X$ in $(X^X)^{\simeq}$. Composing with the canonical map $X^X \to (\mathcal{E}^X)^{(\mathcal{E}^X)}$ we conclude that $(sp)^*$ is homotopy equivalent to $1_{\mathcal{E}^X}$ in $[(\mathcal{E}^X)^{(\mathcal{E}^X)}]^{\simeq}$. □

Proposition 3.4.3. *An inner anodyne map is a Joyal equivalence.*

Proof. Let $X \xrightarrow{i} Y$ be an inner anodyne map of simplicial sets. For any ∞-category $\mathcal{E}$ the induced map $\mathcal{E}^X \xrightarrow{i^*} \mathcal{E}^Y$ is a trivial fibration due to proposition 2.3.3. , which by the previous proposition is a Joyal equivalence. □

Theorem 3.4.4. *Every simplicial set X is Joyal equivalent to an ∞-category.*

Proof. Using small object argument the map $X \to \Delta^0$ can be factored into an inner fibration followed by an inner anodyne map which is Joyal equivalence. Thus X is Joyal equivalent to the factoring object. $\square$

We will show that in fact there is a model structure in simplicial sets (with marking though) where the Joyal equivalence play the role of weak equivalence and the fibrant objects are precisely the ∞-categories. Intuitively the idea of Joyal equivalence is a replacement of the usual notion of 'equivalences of categories' for ∞-categories. The following result is expected as usual.

Definition 3.4.4. *An ∞-functor $\mathcal{C} \xrightarrow{f} \mathcal{D}$ is called essentially surjective if any object x in $\mathcal{D}$ is equivalent to $f(y)$ for some y in $\mathcal{C}$.*

Theorem 3.4.5. *An ∞-functor is a Joyal equivalence iff it is fully faithful and essentially surjective.*

Before we give a proof, let us give some more equivalent criteria for Joyal equivalence.

Proposition 3.4.6. *Let $\mathcal{C} \xrightarrow{p} \mathcal{D}$ be an ∞-functor$\dot{T}$he following conditions are equivalent.*

1. *p is a Joyal equivalence.*
2. *$(\mathcal{C}^X)^{\simeq} \xrightarrow{p_*^{\simeq}} (\mathcal{D}^X)^{\simeq}$ is a homotopy equivalence of kan complexes for any simplicial set X.*
3. *$(\mathcal{C}^{\Delta^n})^{\simeq} \xrightarrow{p_*^{\simeq}} (\mathcal{D}^{\Delta^n})^{\simeq}$ is a homotopy equivalence of kan complexes for n=0,1.*

Proof. $1 \implies 2 \implies 3$ is obvious by definition of Joyal equivalence and its stability under taking groupoid core. We prove the other direction namely $3 \implies 2 \implies 1$.

$3 \implies 2$. We first prove the case for n-spines. For this observe that $n = 1$ case is given since $I^1 = \Delta^1$. Let this be the case for $n = m$. Consider the pushout,

$$\begin{tikzcd} \Delta^0 \arrow[r] \arrow[d] & I^m \arrow[d] \\ \Delta^1 \arrow[r] & I^{m+1} \end{tikzcd}$$

Applying the functor $(\mathcal{C}^{(-)})^{\simeq}$ we get a pull back diagram of kan complexes

$$\begin{tikzcd}
(\mathcal{C}^{I^{m+1}})^{\simeq} \arrow[r] \arrow[d] & (\mathcal{C}^{\Delta^1})^{\simeq} \arrow[d] \\
(\mathcal{C}^{I^m})^{\simeq} \arrow[r] & \mathcal{C}^{\simeq}
\end{tikzcd}$$

The right vertical map is a kan fibration due to theorem 4.2.6, since $\Delta^0 \xrightarrow{\{0\}} \Delta^1$ is a left fibration $\mathcal{C} \to \mathcal{C}^{\Delta^1}$ is a left fibration and thus an isofibration. By functoriality of this functor in $\mathcal{C}$ we get the following diagrams where the left top and bottom horizontal maps are kan fibrations ,and all the vertical maps are homotopy equivalences. (The right and the middle due to condition 3, and the left by hypothesis)

$$\begin{tikzcd}
(\mathcal{C}^{I^m})^{\simeq} \arrow[r] \arrow[d, "heq"] & (\mathcal{C}^{\Delta^1})^{\simeq} \arrow[d, "heq"] & \mathcal{C}^{\simeq} \arrow[l, "kan"'] \arrow[d, "heq"] \\
(\mathcal{D}^{I^m})^{\simeq} \arrow[r] & (\mathcal{D}^{\Delta^1})^{\simeq} & \mathcal{D}^{\simeq} \arrow[l, "kan"]
\end{tikzcd}$$

due to Reedy's lemma we conclude the equivalence on the pullbacks of the top and bottom rows. Now consider the case $X = \Delta^n$. In the below diagram the vertical maps are the equivalences since they are restriction of trivial fibration (as spine inclusion is inner anodyne, applying proposition 2.3.3) to groupoids cores and trivial fibrations are Joyal equivalences.

$$\begin{tikzcd}
(\mathcal{C}^{\Delta^n})^{\simeq} \arrow[r] \arrow[d] & (\mathcal{D}^{\Delta^n})^{\simeq} \arrow[d] \\
(\mathcal{C}^{I^n})^{\simeq} \arrow[r] & (\mathcal{D}^{I^n})^{\simeq}
\end{tikzcd}$$

The bottom map is also an equivalence by the previous step. By 2 out of 3 property for equivalence we conclude that the top map is also an equivalence. Now let X be a 0-dimensional simplicial set. The claim is true in that case, since the map in consideration is given by $\prod_{x \in X_0} \mathcal{C} \xrightarrow{\prod p_*} \prod_{x \in X_0} \mathcal{D}$, since $X = sk_0(X) = \coprod_{x \in X_0} \Delta^0$. And this is a homotopy equivalence as each component is a homotopy equivalence by condition 3. Now for any positive n assume that the claim is true for all $n-1$ dimensional simplicial sets. The

pushout diagram on the left induce the pullback diagram on the right.

$$\begin{tikzcd} \coprod \partial\Delta^n \arrow[r] \arrow[d] & sk_{n-1}(K) \arrow[d] \\ \coprod \Delta^n \arrow[r] & sk_n(K) \end{tikzcd} \qquad \begin{tikzcd} (\mathcal{C}^X)^{\simeq} = (\mathcal{C}^{sk_n(X)})^{\simeq} \arrow[r] \arrow[d] & \prod(\mathcal{C}^{\partial\Delta^n})^{\simeq} \arrow[d] \\ (\mathcal{C}^{sk_{n-1}(K)})^{\simeq} \arrow[r] & (\mathcal{C}^{\Delta^n})^{\simeq} \end{tikzcd} \tag{3.12}$$

Once again we have the following diagram where all the vertical maps equivalences

$$\begin{tikzcd} (\mathcal{C}^{sk_{n-1}(K)})^{\simeq} \arrow[r] \arrow[d, "heq"] & (\mathcal{C}^{\Delta^n})^{\simeq} \arrow[d, "heq"] & \prod(\mathcal{C}^{\partial\Delta^n})^{\simeq} \arrow[l, "kan"'] \arrow[d, "heq"] \\ \mathcal{D}^{sk_{n-1}(K)})^{\simeq} \arrow[r] & (\mathcal{D}^{\Delta^n})^{\simeq} & \prod(\mathcal{D}^{\partial\Delta^n})^{\simeq} \arrow[l, "kan"] \end{tikzcd}$$

(the left and right by induction hypothesis and the middle one by the previous step) and the right top and right bottom maps are kan fibrations (the reason is the same as earlier) and we are done by Reedy's lemma producing the case for n.

Now consider an arbitrary simplicial set $X = \varinjlim_{n<\omega} sk_n(X)$. We have $(\mathcal{C}^X)^{\simeq} = \varprojlim_{n<\omega}(\mathcal{C}^{sk_n(X)})^{\simeq}$ and the following diagram of kan fibration and homotopy equivalences.

$$\begin{tikzcd} \dots \arrow[r, "kan"] & (\mathcal{C}^{sk_{n+1}(K)})^{\simeq} \arrow[r, "kan"] \arrow[d, "heq"] & (\mathcal{C}^{sk_n(K)})^{\simeq} \arrow[r, "kan"] \arrow[d, "heq"] & (\mathcal{C}^{sk_{n-1}(K)})^{\simeq} \arrow[r, "kan"] \arrow[d, "heq"] & \dots \\ \dots \arrow[r, "kan"] & (\mathcal{D}^{sk_{n+1}(K)})^{\simeq} \arrow[r, "kan"] & (\mathcal{D}^{sk_n(K)})^{\simeq} \arrow[r, "kan"] & \mathcal{D}^{sk_{n-1}(K)})^{\simeq} \arrow[r, "kan"] & \dots \end{tikzcd}$$

Thus we get a long diagram of homotopy groups where all the vertical maps are isomorphisms:

$$\begin{tikzcd} \dots \arrow[r] & \pi_i((\mathcal{C}^{sk_{n+1}(X)})^{\simeq}) \arrow[r] \arrow[d, "\cong"] & \pi_i((\mathcal{C}^{sk_n(X)})^{\simeq}) \arrow[r] \arrow[d, "\cong"] & \pi_i((\mathcal{C}^{sk_{n-1}(X)})^{\simeq}) \arrow[r] \arrow[d, "\cong"] & \dots \\ \dots \arrow[r] & \pi_i((\mathcal{D}^{sk_{n+1}(X)})^{\simeq}) \arrow[r] & \pi_i((\mathcal{D}^{sk_n(X)})^{\simeq}) \arrow[r] & \pi_i((\mathcal{D}^{sk_{n-1}(X)})^{\simeq}) \arrow[r] & \dots \end{tikzcd}$$

Clearly this provides an isomorphism $\varprojlim_{n<\omega} \pi_i((\mathcal{C}^X)^{\simeq}) \xrightarrow{\cong} \varprojlim_{n<\omega} \pi_i((\mathcal{D}^X)^{\simeq})$. We have the following diagram of short exact sequence where the right vertical map is an isomorphism since the homotopy group functor preserve transfinite homotopy limit.

$$\begin{tikzcd}
0 \arrow[r] & \underset{n<\omega}{\underleftarrow{\mathrm{holim}}}\, \pi_{i+1}((\mathcal{C}^X)^{\simeq}) \arrow[r] \arrow[d] & \pi_i(\underset{n<\omega}{\underleftarrow{\lim}}(\mathcal{C}^X)^{\simeq}) \arrow[r] \arrow[d] & \underset{n<\omega}{\underleftarrow{\lim}}\, \pi_i((\mathcal{C}^X)^{\simeq}) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \underset{n<\omega}{\underleftarrow{\mathrm{holim}}}\, \pi_{i+1}((\mathcal{D}^X)^{\simeq}) \arrow[r] & \pi_i(\underset{n<\omega}{\underleftarrow{\lim}}(\mathcal{D}^X)^{\simeq}) \arrow[r] & \underset{n<\omega}{\underleftarrow{\lim}}\, \pi_i((\mathcal{D}^X)^{\simeq}) \arrow[r] & 0
\end{tikzcd}$$

By 3-lemma the middle vertical map is an isomorphism. Since the corresponding objects are kan complexes, by Whitehead's theorem $(\mathcal{C}^X)^{\simeq} \xrightarrow{p_*^{\simeq}} (\mathcal{D}^X)^{\simeq}$ is a homotopy equivalence (the case $n = 0$ follows similarly in sets).

It remains to prove 2 $\implies$ 1. For this purpose choose $X = \mathcal{D}$. The above homotopy equivalence now reads as $(\mathcal{C}^{\mathcal{D}})^{\simeq} \xrightarrow{p_*^{\simeq}} (\mathcal{D}^{\mathcal{D}})^{\simeq}$. Since this induce bijection on π_0 and $1_{\mathcal{D}} \in \mathcal{D}^{\mathcal{D}}$, there exists an object $g \in (\mathcal{C}^{\mathcal{D}})^{\simeq}$ such that $p_*(g) \simeq 1_{\mathcal{D}}$ in the ∞-category $(\mathcal{D}^{\mathcal{D}})^{\simeq}$. That is there is an ∞-functor $\mathcal{D} \xrightarrow{g} \mathcal{C}$ such that pg is naturally equivalent to $1_{\mathcal{D}}$. Similarly putting $X = \mathcal{C}$ we have a hompotopy equivalence $(\mathcal{C}^{\mathcal{C}})^{\simeq} \xrightarrow{p_*^{\simeq}} (\mathcal{D}^{\mathcal{C}})^{\simeq}$. Now for $gp \in (\mathcal{C}^{\mathcal{C}})^{\simeq}$ we see that $p_*(gp) = pgp \simeq p = p_*(1_{\mathcal{C}})$. Since p_* is bijective on π_0 we get $gp \simeq 1_{\mathcal{C}}$ in $(\mathcal{C}^{\mathcal{C}})^{\simeq}$, that is gp is naturally equivalent to $1_{\mathcal{C}}$. □

Proposition 3.4.7. *Joyal equivalences are full and faithful.*

Proof. Let $\mathcal{C} \xrightarrow{f} \mathcal{D}$ be a Joyal equivalence. By definition there exists an ∞-functor $\mathcal{D} \xrightarrow{g} \mathcal{C}$ such that $gf \overset{\alpha}{\sim} 1_{\mathcal{C}}$ and $fg \overset{\beta}{\sim} 1_{\mathcal{D}}$ such that α, β are natural equivalences. Then for any object x in $\mathcal{D}$, $\beta_x : x \to f(gx)$ is an equivalence in $\mathcal{D}$ where $gx \in \mathcal{C}$. Thus f is essentially surjective. Moreover in the following sequence :

$$\mathcal{C}_1^{\infty}(x, y) \xrightarrow{f_{x,y}} \mathcal{D}_1^{\infty}(fx, fy) \xrightarrow{g_{fx,fy}} \mathcal{C}_1^{\infty}(gfx, gfy) \xrightarrow{f_{gfx,gfy}} \mathcal{D}_1^{\infty}(fgfx, fgfy)$$

the composite of the first two maps is homotopic to identity as α is a natural equivalence and so is the composite of the last two maps. Therefore the middle map has both a left and right homotopy inverse. Whence the middle map is a homotopy equivalence and hence so is the right inverse $f_{x,y}$, which proves the full and faithfulness of f. □

SubTheorem 3.4.8. *Fully faithful and essentially surjective ∞-functors between kan complexes are Joyal equivalence (which is just a homotopy equivalence of kan complexes as mentioned in the definition).*

Proof. Consider a fully faithful and essentially surjective ∞-functor $\mathcal{K} \xrightarrow{p} \mathcal{L}$ of kan complexes. To prove that this is a homotopy equivalence by Whitehead's theorem, it suffices to prove that it is a weak equivalence of simplicial sets. Let x be an arbitrary object in $\mathcal{K}$ and let $y = fx$. Define $Dom_{\mathcal{K}}(x)$ as the ∞-category of maps with domain x in the ∞-category $\mathcal{K}$ as the following pull back diagram.

$$\begin{tikzcd} Dom_{\mathcal{K}}(x) \arrow[r] \arrow[d] & \mathcal{K}^{\Delta^1} \arrow[d, "d_1"] \\ \Delta^0 \arrow[r, "x"] & \mathcal{K} \end{tikzcd}$$

The right hand square in the below diagram is still a pullback diagram,

$$\begin{tikzcd} \mathcal{K}_1^{\infty}(x,x) \arrow[r] \arrow[d] & Dom_{\mathcal{K}}(x) \arrow[r] \arrow[d] & \mathcal{K}^{\Delta^1} \arrow[d, "(d_1,d_0)"] \\ \Delta^0 \arrow[r, "(1,x)"] & \Delta^0 \times \mathcal{K} \arrow[r, "x\times 1_{\mathcal{K}}"] & \mathcal{K} \times \mathcal{K} \end{tikzcd}$$

where the middle map is induced by d_0 on $\mathcal{K}$.

By pasting lemma the left one is therefore a pull back diagram, since the whole diagram is a pullback by definition. Now observe the right most vertical map is induced by the boundary inclusion $\partial\Delta^1 \hookrightarrow \Delta^1$ on the terminal map $\mathcal{K} \to \Delta^0$, which is a kan fibration. By Proposition 2.3.3 the right vertical map is therefore a kan fibration. But kan fibrations are stable under pull back. Thus we obtain that the middle map is also a Kan fibration. Note that the same argument applied to the first diagram defining $Dom_{\mathcal{K}}(x)$ implies that the right map d_1 is a trivial fibration which implies by stability of trivial fibration under pull back that $Dom_{\mathcal{K}}(x)$ is a contractible kan complex. Now we have produced a kan fiber sequence of kan complexes and thus a map of kan fiber sequences as below.

$$\begin{tikzcd} \mathcal{K}_1^{\infty}(x,x) \arrow[r] \arrow[d] & Dom_{\mathcal{K}}(x) \arrow[r] \arrow[d] & \mathcal{K} \arrow[d] \\ \mathcal{L}_1^{\infty}(y,y) \arrow[r] & Dom_{\mathcal{L}}(y) \arrow[r] & \mathcal{L} \end{tikzcd}$$

Since the middle objects are contractible we have in the following commutative diagram on homotopy groups (use long exact sequence of homotopy groups for a kan fiber sequence of

kan complexes for $n \geq 1$) that the horizontal maps are isomorphisms.

$$\begin{tikzcd}
\pi_n(\mathcal{K}, x) \arrow[r, "\partial"] \arrow[d] & \pi_{n-1}(\mathcal{K}_1^\infty(x,x), 1_x) \arrow[d] \\
\pi_n(\mathcal{L}, fx) \arrow[r, "\partial"] & \pi_{n-1}(\mathcal{L}_1^\infty(fx, fx), 1_{fx})
\end{tikzcd}$$

But the right vertical map is also an isomorphism since $\mathcal{K}_1^\infty(x,x) \xrightarrow{f_{x,x}} \mathcal{L}_1^\infty(fx, fx)$ is a homotopy equivalence due to f being fully faithful. By commutative of the square we conclude that f induce isomorphism on π_n , $(n \geq 1)$ for any base point x. The remaining case is $n = 0$ which is just essential surjectivity. $\square$

Proposition 3.4.9. *Fully faithful and essentially surjective ∞-functors are Joyal equivalences.*

Proof. Let $\mathcal{C} \xrightarrow{p} \mathcal{D}$ be a fully faithful and essentially surjective ∞-functorẆe show that condition of proposition 3.4.6 holds for p.

1. **n=0**
   In this case the map of interest is $\mathcal{C}^\simeq \xrightarrow{p^\simeq} \mathcal{D}^\simeq$. We claim that this is also fully faithful and essentially surjective. Since domain and target of this map are kan complexes, by the previous subtheorem 3.4.8 we will be done. As objects of the groupoid cores are the same as that of the corresponding ∞-categories the essential surjectivity of $p^\simeq$ is immediate from essential surjectivity of p. For fully faithfulness observe that the following is a pull back diagram.

$$\begin{tikzcd}
\mathcal{C}^\simeq \arrow[r, hook] \arrow[d] & \mathcal{C} \arrow[d] \\
\mathcal{D}^\simeq \arrow[r, hook] & \mathcal{D}
\end{tikzcd} \tag{3.13}$$

   Since pullbacks are stable under product and exponential, in the following diagram the right square is a pull back by replacing the ∞-categories by their corresponding power objects, whereas the whole outer diagram is a pullback by applying $(_)^{\Delta^1}$ to the above

pullback diagram. Therefore the left square of is also a pullback.

$$\begin{tikzcd}
(\mathcal{C}^{\simeq})^{\Delta^1} \arrow[r] \arrow[d] & (\mathcal{C}^{\Delta^1})^{\simeq} \arrow[r] \arrow[d] & \mathcal{C}^{\simeq} \times \mathcal{C}^{\simeq} \arrow[d] \\
(\mathcal{D}^{\simeq})^{\Delta^1} \arrow[r] & (\mathcal{D}^{\Delta^1})^{\simeq} \arrow[r] & \mathcal{D}^{\simeq} \times \mathcal{D}^{\simeq}
\end{tikzcd}$$

Therefore in the following diagram the left front face is a pullback. Let $x, y \in \mathcal{C}^{\simeq}$ be two objects, and consider fibers on the top and bottom faces in the below diagram. By pasting lemma applied to the top and bottom faces we get that the top left and bottom left faces are pull backs as well (since the fact that the right and the whole outer diagrams on top and bottom faces are pullbacks by definition and that $\mathcal{C}^{\simeq} \times \mathcal{C}^{\simeq} = (\mathcal{C} \times \mathcal{C})^{\simeq}$ $\mathcal{D}^{\simeq} \times \mathcal{D}^{\simeq} = (\mathcal{D} \times \mathcal{D})^{\simeq}$)

Now looking at the left side cube in the same diagram we find that the top bottom and front faces are pull backs. By pasting lemma of pull backs we conclude that the behind face is also a pullback. Since the middle map in the back face is a homotopy equivalence we deduce that so is the left most map in the back face as the left back bottom map $(\mathcal{D}^{\simeq})_1^{\infty}(px, py) \to \mathcal{D}_1^{\infty}(px, py)$ is a kan fibration, being pull back of $(\mathcal{D}^{\simeq})^{\Delta^1} \to (\mathcal{D}^{\Delta^1})^{\simeq}$. Whence $p^{\simeq}$ is indeed fully faithful.(The proof that the map $(\mathcal{D}^{\simeq})^{\Delta^1} \to (\mathcal{D}^{\Delta^1})^{\simeq}$ is a kan fibration is straightforward. The composite in the front bottom row is conservative due to proposition 4.2.8 which implies the conservativity of the first map $(\mathcal{D}^{\simeq})^{\Delta^1} \to (\mathcal{D}^{\Delta^1})^{\simeq}$. We allude to theorem 4.2.4 since the condition of being equivalence on the images of ϕs are redundant for the target is an ∞-groupoid)

$$\begin{tikzcd}
& (\mathcal{C}^{\simeq})_1^{\infty}(x,y) \arrow[rr] \arrow[dl] \arrow[dd, "p^{\simeq}_{x,y}"] & & \mathcal{C}_1^{\infty}(x,y) \arrow[rr, "p_{x,y}"] \arrow[dl] \arrow[dd] & & \Delta^0 \arrow[dl, "(x,y)"] \arrow[dd] \\
(\mathcal{C}^{\simeq})^{\Delta^1} \arrow[rr] \arrow[dd] & & (\mathcal{C}^{\Delta^1})^{\simeq} \arrow[rr] \arrow[dd] & & \mathcal{C}^{\simeq} \times \mathcal{C}^{\simeq} \arrow[dd] & \\
& (\mathcal{D}^{\simeq})_1^{\infty}(px,py) \arrow[rr] \arrow[dl] & & \mathcal{D}_1^{\infty}(px,py) \arrow[rr] \arrow[dl] & & \Delta^0 \arrow[dl, "(px,py)"] \\
(\mathcal{D}^{\simeq})^{\Delta^1} \arrow[rr] & & (\mathcal{D}^{\Delta^1})^{\simeq} \arrow[rr] & & \mathcal{D}^{\simeq} \times \mathcal{D}^{\simeq} &
\end{tikzcd}$$

2. **n=1**

The top and bottom right faces in the following diagram are pull backs by inspec-

tion.Since the middle maps into the page maps are restrictions of the isofibrations induced by the boundary inclusion(theorem 4.2.12) to the the groupoid cores, they are kan fibrations. The behind middle vertical map is a homotopy equivalence by the previous step. To show that the middle front vertical map is a homotopy equivalence it suffices to show that this is the case fiber-wise.

$$\begin{tikzcd}
 & \Delta^0 \arrow[rr, "{(x,y)}"] \arrow[dd] & & \mathcal{C}^{\simeq} \times \mathcal{C}^{\simeq} \arrow[rr] \arrow[dd] & & \mathcal{C} \times \mathcal{C} \arrow[dd] \\
\mathcal{C}_1^{\infty}(x,y) \arrow[ur] \arrow[rr] \arrow[dd, "p_{x,y}"] & & (\mathcal{C}^{\Delta^1})^{\simeq} \arrow[ur] \arrow[rr] \arrow[dd] & & \mathcal{C}^{\Delta^1} \arrow[ur] \arrow[dd] & \\
 & \Delta^0 \arrow[rr, "{(px,py)}"] & & \mathcal{D}^{\simeq} \times \mathcal{D}^{\simeq} \arrow[rr] & & \mathcal{D} \times \mathcal{D} \\
\mathcal{D}_1^{\infty}(px,py) \arrow[ur] \arrow[rr] & & (\mathcal{D}^{\Delta^1})^{\simeq} \arrow[ur] \arrow[rr] & & \mathcal{D}^{\Delta^1} \arrow[ur] &
\end{tikzcd}$$

But the fibers are just the ∞homs (by pasting lemma of pullbacks) and the corresponding map on fibers are just $p_{x,y}$ induced by p on ∞homs, which are homotopy equivalences since p is fully faithful.

□

Example 1. ***Equivalence of abstract and concrete Spaces***

The two different ∞-categories of spaces as mentioned in example 2.5.1 and 2.5.2 are Joyal equivalent. We know that the category of Kan complexes and CW complexes are equivalent not only as ordinary categories but also as simplicially enriched categories. Their homotopy coherent nerves are therefore Joyal equivalent.

3.5 Localization of ∞ -categories

The theory of ∞-categories we present here is a model for $(\infty, 1)$-categories since the corresponding ∞homs are kan complexes or ∞-groupoids. That is in an ∞-category all k-morphisms above 1 are invertible. Therefore it makes sense to talk about inverting only

1-morphisms in ∞-categories. Of course here inverting a morphism would mean that it is made an equivalence, since in ∞-categories equivalences are best replacement for isomorphisms. In the following section we will talk about the idea of "Localizations" of an ∞-category along a set of 1-morphisms. We will talk about this more in the next chapter, since the language required for this is better suited by the concepts there.

3.5.1 Dwyer Kan Localization

Given a set Σ of morphisms in an ordinary category $\mathscr{C}$ the definition of localization of $\mathscr{C}$ along Σ, denoted $\mathscr{C}[\Sigma^{-1}]$ is given by a natural isomorphism : $Cat(\mathscr{C}, \mathscr{D})^{\Sigma} \xrightarrow{\cong} Cat(\mathscr{C}[\Sigma^{-1}], \mathscr{D})$ (natural in $\mathscr{D}$), where the category $Cat(\mathscr{C}, \mathscr{D})^{\Sigma}$ is the full subcategory of $Cat(\mathscr{C}, \mathscr{D})$ on functors which sends all the morphisms in Σ are sent to isomorphisms in $\mathscr{D}$.

Definition 3.5.1. *Let Σ be a set of 1-morphisms in an ∞-category $\mathcal{C}$. The localization of $\mathcal{C}$ along Σ , if it exists is an ∞-category $\mathcal{C}[\Sigma^{-1}]$ such that for any ∞-category $\mathcal{D}$ the full subcategory (denoted $\mathcal{D}^{\mathcal{C}}_{\Sigma}$) of $\mathcal{D}^{\mathcal{C}}$ on ∞-functors which take all morphisms of Σ to equivalences of $\mathcal{D}$, is Joyal equivalent to the ∞-category $\mathcal{D}^{\mathcal{C}[\Sigma^{-1}]}$. More precisely a localization is an ∞-functor $\mathcal{C} \xrightarrow{l} \mathcal{C}[\Sigma^{-1}]$ such that for any ∞-category $\mathcal{D}$ the restriction functor $\mathcal{D}^{\mathcal{C}[\Sigma^{-1}]} \xrightarrow{l^*} \mathcal{D}^{\mathcal{C}}$ is fully faithful and its essential image consists of those ∞-functors $\mathcal{C} \to \mathcal{D}$ which takes Σ to equivalences in $\mathcal{D}$.*

Proposition 3.5.1. *A localization if it exists is unique up to a Joyal equivalence.*

Proof. Let $\mathcal{L}_1, \mathcal{L}_2$ be two localization. By definition, for any auxiliary ∞-category $\mathcal{D}$, there are Joyal equivalences, $\mathcal{D}^{\mathcal{L}_1} \cong \mathcal{D}^{\mathcal{C}}_{\Sigma} \cong \mathcal{D}^{\mathcal{L}_2}$. Since $\mathcal{D}$ is arbitrary one deduces map $\mathcal{L}_1 \to \mathcal{L}_2$ by putting $\mathcal{D} = \mathcal{L}_2$. Then by theorem 3.4.1, it follows that $\mathcal{L}_1 \cong \mathcal{L}_2$. $\square$

Theorem 3.5.2. *A localization of an ∞-category along any set of morphisms always exists.*

An easy proof of the above theorem can be deduced from the following result by taking coproduct along the set S of the localization $\Delta^1 \to J$.

Proposition 3.5.3. *The localization of Δ^1 along the single morphism $0 \to 1$ is the nerve of the category $\boldsymbol{J}$ with two objects and just a nontrivial isomorphism between them. We denote this nerve as J.*

Proof. One has to show that for any ∞-category $\mathcal{C}$ the the canonical map $\mathcal{C}^J \xrightarrow{i^*} \mathcal{C}^{\Delta^1}$ is fully faithful, since the essential image is exactly those maps which takes the map $0 \to 1$ in Δ^1 to equivalences in $\mathcal{C}$ already.

$$\begin{tikzcd}
\mathcal{C}^{\mathcal{J}}{}_1^{\infty}(x,y) \arrow[r] \arrow[d] & \mathcal{C}^{\Delta^1 \times J} \arrow[r] \arrow[d, "(d_1,d_0)"] & \mathcal{C}^{\Delta^1 \times \Delta^1} \arrow[d, "(d_1,d_0)"] \\
\Delta^0 \arrow[r, "(x,y)"] & \mathcal{C}^J \times \mathcal{C}^J \arrow[r, "i^* \times i^*"'] & \mathcal{C}^{\Delta^1} \times \mathcal{C}^{\Delta^1}
\end{tikzcd}$$

where the right hand diagram is really a pull back as it is induced by the push out $J \times \Delta^1 = (\Delta^1 \times \Delta^1) \coprod_{\Delta^1 \coprod \Delta^1} (J \coprod J)$ by applying the exponential $\mathcal{C}^{(-)}$. Whence for any x, y, by pasting lemma of pull backs $(\mathcal{C}^J)_1^{\infty}(x,y) \cong (\mathcal{C}^{\Delta^1})_1^{\infty}(x,y)$ (isomorphism!). $\square$

Proof. (of theorem 3.5.2) Consider the pushout on the left. For any arbitrary ∞-category $\mathcal{D}$, we have the right diagram as a pull back, where the bottom map is a full faithful from the previous proof, and the right vertical map is an isofibration due to 3.2.13, we deduce the full and faithfulness of the top horizontal map $\mathcal{D}^{\mathcal{L}} \to \mathcal{D}^{\mathcal{C}}$.

$$\begin{tikzcd}
\coprod_{f \in \Sigma} \Delta^1 \arrow[r] \arrow[d] & \mathcal{C} \arrow[d] \\
\coprod_{f \in \Sigma} J \arrow[r] & \mathcal{L}
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathcal{D}^{\mathcal{L}} \arrow[r] \arrow[d] & \mathcal{D}^{\mathcal{C}} \arrow[d] \\
\prod_{f \in \Sigma} \mathcal{D}^J \arrow[r] & \prod_{f \in \Sigma} \mathcal{D}^{\Delta^1}
\end{tikzcd} \tag{3.14}$$

The deal with essential image is clear from the definition of $\mathcal{L}$. Therefore $\mathcal{L}$ is indeed a localization of $\mathcal{C}$ along Σ. $\square$

3.5.2 Examples

Example 2. ***Localization of an Ordinary category***

Let $\mathscr{C}$ be an ordinary category and Σ a set of morphisms in $\mathcal{C}$. Then the localization of the nerve of $\mathscr{C}$ along Σ, $\mathcal{L} = N(\mathscr{C})[\Sigma^{-1}]$ exists, as proved in the previous step. From the description of $\mathcal{L}$, by applying the functor h (having a right adjoint), we find that $h(N(\mathscr{C})[\Sigma^{-1}]) = \mathscr{C}[\Sigma^{-1}]$. That is one can produce the ordinary categorical localization of

an ordinary category by taking the homotopy category of the ∞ categorical localization of its nerve. However it is not true always that $N(\mathscr{C})[\Sigma^{-1}] = N(\mathscr{C}[\Sigma^{-1}])$, as ∞-categories.

Example 3. ***Derived ∞-category of an Abelian category***

Consider the homotopy ∞-category as described in example 2.5.1. Let Σ be the set of all quasi-isomorphisms in this category. The Dwyer-Kan localization exists and is called the Derived ∞-category of $\mathscr{A}$. This is clearly an ∞-categorical generalization of the ordinary derived category. Now one might wonder why is such a complication even needed. The answer lies in the fact the ordinary derived category is very rough in the sense that once quasi-isomorphisms are inverted, the information about the homotopies is lost in the ordinary derived category. However this information is retained in the higher simplices in the derived ∞-category $\mathfrak{D}(\mathscr{A}) = \mathfrak{K}(\mathscr{A})[qi^{-1}]$. Note that the the homotopy category of $\mathfrak{D}(\mathscr{A})$ is precisely the ordinary derived category.

Chapter 4

J-model and Marked model structure on Simplicial sets

When dealing with localization of ∞-categories we have encountered that the one side of the equivalence is given by maps which takes certain morphisms to certain morphisms as we want. Thus we can think of such set of morphisms as markings of the corresponding ∞-categories. And maps are functors which preserve them. In this chapter we elaborate this idea to some detail and reformulate localization via a different model structure. We will also produce a model structure in simplicial sets where Joyal equivalence play the role of weak equivalences.The fibrant objects are precisely the ∞-categories in both the model structure. And finally we will talk about the Strengthening Un-strengthening method, which play an essential role in further development of the theory.

4.1 Jmodel structure

Among all the objects in a simplicially enriched category the fibrant objects are nice in some sense and has easily derivable properties. So far what we have is that in the usual Quillen model structure for simplicial sets the fibrant objects are kan complexes and are therefore ∞-categories. However since all ∞-categories are not kan complexes, properties of ∞-categories are not easily understandable under this model structure. Since every simplicial set is Joyal equivalent to an ∞-category via a monomorphism (inner anodyne to be precise) we can

still hope for an Injective Model structure on simplicial sets with Joyal equivalence as weak equivalences, where the fibrant objects are exactly the ∞-categories as we will to have. The following lemma motivates us further.

Lemma 4.1.1. *Joyal equivalence has 2-out-of-3 property and are closed under retracts.*

Proof. Since homotopy equivalences has 2-out-of-3 property(moreover between kan complexes they are just weak homotopy equivalence), due to proposition 3.4.6 the claim follows for any ∞-functorṠince Joyal equivalence of general simplicial sets is defined via (induced) Joyal equivalence of ∞-categories the claim follows for general simplicial set map. For retracts it is similarly enough to check the case for ∞-categories. Let f be a retract of g where g is a Joyal equivalence:

$$\begin{tikzcd}
\mathcal{A} \arrow[r, "a"] \arrow[d, "f"] & \mathcal{X} \arrow[r, "x"] \arrow[d, "g"] & \mathcal{A} \arrow[d, "f"] \\
\mathcal{B} \arrow[r, "k"'] & \mathcal{Y} \arrow[r, "b"] & \mathcal{B}
\end{tikzcd}$$

If $\mathcal{Y} \xrightarrow{h} \mathcal{X}$ is a Joyal inverse for g then xhk is a Joyal inverse for f. Note that indeed, $xhkf = xhga \simeq xa = 1_{\mathcal{A}}$ and $fxhk = bghk \simeq bk = 1_{\mathcal{B}}$, and the homotopies are equivalences automatically. $\square$

By the above lemma we see that Joyal equivalences are pretty good in nature to be weak equivalences. Now if we require an injective model structure on simplicial sets, we must make sure that the class of acyclic fibrations do not change as it is entirely determined by cofibrations. The situation is revealed in case of ∞-categories and provides a hint for the class of fibration (at least for ∞-categories).

Proposition 4.1.2. *An ∞-functor is a trivial fibration in the Quillen model structure iff it is a Joyal equivalence and an isofibration.*

Proof. That trivial fibrations are Joyal equivalence and isofibrations is clear from proposition 3.4.2 and 3.2.2 respectively. It remains to prove the converse. A detailed proof can be found in M. Land[9]. $\square$

The next proposition gives a view towards the proposed model structure by proposing a functorial factorization.

Proposition 4.1.3. *There is a functorial factorization (α, β) of ∞-functors such that for any ∞-functor $\mathcal{C} \xrightarrow{p} \mathcal{D}$ of ∞-categories the map $\alpha(p)$ is an isofibration and $\beta(p)$ is a Joyal equivalence which is a monomorphism.*

Proof. Consider the pull back,

$$\begin{tikzcd} P(p) \arrow[r, "d_1'"'] \arrow[d, "p'"] & \mathcal{C} \arrow[d, "p"] \\ \mathcal{D}^J \arrow[r, "d_1"'] & \mathcal{D} \end{tikzcd} \qquad \begin{tikzcd} P(p) \arrow[r] \arrow[d, "p'"] & \mathcal{C} \times \mathcal{D} \arrow[r, "pr_{\mathcal{C}}"] \arrow[d, "p \times 1"] & \mathcal{C} \arrow[d, "p"] \\ \mathcal{D}^J \arrow[r] & \mathcal{D} \times \mathcal{D} \arrow[r] & \mathcal{D} \end{tikzcd} \tag{4.1}$$

The map $\mathcal{C} \times J \to \mathcal{C} \to \mathcal{D}$, by adjunction induce a map $\mathcal{C} \xrightarrow{f} \mathcal{D}^J$. Clearly $d_1 f = p$. Thus by universality we get a map $\mathcal{C} \xrightarrow{\beta(p)} P(p)$ such that $d_1'\beta(p) = 1_{\mathcal{C}}$ and $p'\beta(p) = f$. Let $\alpha(p) = pd_1' = d_1 p'$. Note that $\alpha(p) \circ \beta(p) = p$. That $\beta(p)$ is a monomorphism is clear from the identity, $d_1'\beta(p) = 1_{\mathcal{C}}$. Since $\{0\} \to J$ is clearly fully faithful (for $J_1^{\infty}(\{0\}, \{0\}) = \Delta^0 = {\Delta^0}_1^{\infty}(\{0\}, \{0\})$) and essentially surjective (as J is an ∞-groupoid) it is a Joyal equivalence. Therefore d_1 is a joyal equivalence. Moreover since $\mathcal{D} \to \Delta^0$ is an isofibration, therefore by theorem 4.1.2 d_1 is an isofibration as well. Hence d_1 is a trivial fibration. Since trivial fibration are stable under pull backs, d_1' is also a trivial fibration and hence a Joyal equivalence. Due to the formulae $d_1'\beta(p) = 1_{\mathcal{C}}$ we conclude that that $\beta(p)$ is a Joyal equivalence by 2-out-of-3 property. For $\alpha(p)$, note that the map

$$\mathcal{D}^J \longrightarrow \mathcal{D} \times \mathcal{D}$$

in the left diagram is induced by the monomorphism $\partial\Delta^1 \to J$ which is a bijection on zero simplices, and is thus an isofibration by theorem 3.2.12. The second map is an isofibration by inspection. Consequently the pull back $P(p) \to \mathcal{C} \times \mathcal{D}$ is an isofibrtion and therefore so is the composite $P(p) \longrightarrow \mathcal{C} \times \mathcal{D} \xrightarrow{pr_{\mathcal{D}}} \mathcal{D}$ which is equal to $\alpha(p)$. $\square$

Theorem 4.1.4. *There is a model structure on simplicial sets where cofibrations are monomorphisms, weak equivalences are Joyal equivalence.*

Proof. We will use Jeff smith's theorem to prove this. The fact that Simplicial sets is locally presentable is well known. Let I be the set of boundary inclusions $I = \{\partial\Delta^n \hookrightarrow \Delta^n |\ n \geq 0$ and $\mathbb{JE}$ be the class of Joyal equivalence of simplicial sets .The solution set condition of $\mathbb{JE}$

at I can be happily forgotten assuming Vopenka's principle, see for e.g. corollary 4.4 in [10]. The following three conditions remains to be checked.

1. $\mathbb{JE}$ has 2-out-of-3 property.
2. $inj(I) \subset \mathbb{JE}$
3. $cof(I) \cap \mathbb{JE}$ is closed under transfinite composition and pushout.

Note that $cof(I)$ is precisely the class of all monomorphisms. Also, $inj(I)$ is precisely the class of all trivial fibrations in simplicial model structure whence condition 2 follows immediately from proposition 3.4.2. Condition 1 has been proven in lemma 4.1.1 and condition 3 will be proved in parts as proposition 4.1.6 and 4.1.7. Also note that condition 3 is separately true for $cof(I)$ and $\mathbb{JE}$. The case for $cof(I)$ is standard, we will prove the case for Joyal equivalence. □

Proposition 4.1.5. *In the following pull back diagram of ∞-categories if the bottom map i is an isofibration and p is a Joyal equivalence then $i^{-1}(p)$ is a Joyal equivalence. In other words Joyal equivalence of ∞-categories are stable under pull back along isofibrations of ∞-categories.*

Proof. We will use proposition 3.4.6. to prove this. Let X be an arbitrary simplicial set. We wish to show that $(\mathcal{A}^X)^{\simeq} \xrightarrow{i^{-1}(p)^{\simeq}_*} (\mathcal{B}^X)^{\simeq}$ is a homotopy equivalence of kan complexes. Since both powerobject functor and groupoid cores preserve pullbacks the right diagram is also a pullback.

$$\begin{tikzcd} \mathcal{A} \arrow[r, "i^{-1}(p)"] \arrow[d] & \mathcal{B} \arrow[d, "i"] \\ \mathcal{C} \arrow[r, "p"'] & \mathcal{D} \end{tikzcd} \qquad \begin{tikzcd} (\mathcal{A}^X)^{\simeq} \arrow[r, "i^{-1}(p)^{\simeq}_*"] \arrow[d] & (\mathcal{B}^X)^{\simeq} \arrow[d, "i^{\simeq}_*"] \\ (\mathcal{C}^X)^{\simeq} \arrow[r, "p^{\simeq}_*"'] & (\mathcal{D}^X)^{\simeq} \end{tikzcd} \tag{4.2}$$

But since p is given to be a Joyal equivalence already by the same proposition (3.4.6 condition 2) we have that the bottom map of the right square is a homotopy equivalence. On the other hand by theorem 4.1.2 $\mathcal{B}^X \xrightarrow{i_*} \mathcal{D}^X$ is an isofibration. Therefore by theorem 4.2.6 the right vertical map is a kan fibration. By Reedy's lemma applied to the right diagram we find that $i^{-1}(p)^{\simeq}_*$ is indeed a homotopy equivalence. □

Proposition 4.1.6. *Joyal equivalence which are monomorphisms are stable under pushout along arbitrary map.*

Proof. Consider a pushout diagram of simplicial sets on the left. Assume that p is a Joyal equivalence which is also a monomorphism. We want to show that p' is also a Joyal equivalence. That is for any ∞-category $\mathcal{C}$ the induced map p'^* is a Joyal equivalence of ∞-categories.

$$\begin{tikzcd} X \arrow[r, "p"] \arrow[d, "i"] & Y \arrow[d, "i'"] \\ Z \arrow[r, "p'"'] & W \end{tikzcd} \qquad \begin{tikzcd} \mathcal{C}^W \arrow[r, "i'^*"] \arrow[d, "p'^*"'] & \mathcal{C}^Y \arrow[d, "p^*"] \\ \mathcal{C}^Z \arrow[r, "i^*"'] & \mathcal{C}^X \end{tikzcd} \tag{4.3}$$

p^* is a Joyal equivalence by definition. Moreover since p is a monomorphism and $\mathcal{C} \to \Delta^0$ is always an isofibration, consequently p^* is an isofibration due to theorem 4.1.2. Thus p^* is a trivial fibration. Since trivial fibrations are stable under pull backs, p'^* is a trivial fibration as well. So p'^* is a Joyal equivalence by theorem . □

Proposition 4.1.7. *The class of Joyal equivalence is closed under transfinite composition.*

Proof. Once again Joyal equivalence between general simplicial sets are defined via Joyal equivalence between ∞-categories. Therefore it is enough to check that Joyal equivalence of ∞-categories are stable under transfinite limits. But for ∞-categories this condition boils down to the case of homotopy equivalences of kan complexes, which is indeed stable under transfinite limits. □

Theorem 4.1.8. *A map of simplicial sets $\mathbf{X} \xrightarrow{f} \mathcal{C}$, where $\mathcal{C}$ is an ∞-category , is a fibration in the J-model structure iff it is an isofibration. (Note that isofibrations are always inner fibrations for us)*

To prove this theorem we would like to talk about J-anodyne maps which are certain trivial cofibrations in the J-model structure forming a saturated class, though they are not all. The starting point of this, is the observation that the inclusion map $\{0\} \to J$ is a trivial cofibration since it is both a Joyal equivalence and a monomorphism, as described in the previous proof. However it is not an anodyne map. On the other hand every inner anodyne map is a Joyal equivalence and a monomorphism and thus a trivial cofibration in the J-model structure.

Definition 4.1.1. *J-anodyne maps is the saturated class of the the set $S = \{\Lambda^n_j \hookrightarrow \Delta^n, \{0\} \to J \mid 0 < j < n\}$.*

Since the set S is contained in the intersection $\mathbb{JE} \cap Mono$, which is saturated, every J-anodyne map is a Joyal Equivalence and a monomorphism. The following lemmas show that J-anodyne maps are really Joyal trivial cofibration at least for ∞-categories.

Lemma 4.1.9. *Let $X \xrightarrow{f} \mathcal{C}$ be a monomorphism where $\mathcal{C}$ is an ∞-category. Then f is a J-anodyne map iff it is a Joyal equivalence.*

Proof. It remains to prove that if f is a Joyal equivalence then it is J-anodyne. Factor f by small object argument on the set S, as $X \xrightarrow{g} \mathcal{D} \xrightarrow{h} \mathcal{C}$ where g is J-anodyne and h has RLP with respect to J-anodyne maps. Since inner anodyne maps are J-anodyne, g having RLP with respect to J-anodyne maps implies that $\mathcal{D}$ is an ∞-category since $\mathcal{C}$ is. Furthermore both f and g are Joyal equivalence, so is h by 2-out-of-3 property. We get from the following commutative diagram:

$$\begin{tikzcd} X \arrow[r, "g"] \arrow[d, "f"'] & \mathcal{D} \arrow[d, "h"] \\ \mathcal{C} \arrow[r, "1"] \arrow[ur, dashed, "k"] & \mathcal{C} \end{tikzcd}$$

where h has right lifting property with respect to J-anodyne maps imply that it is an isofibration of ∞-categories. As a result h is a (Quillen) trivial fibration due to proposition 5.1.2. But f is a monomorphism as well. Thus the above diagram has a lift, say k. Then f is a retract of g and is thus J-anodyne.

$$\begin{tikzcd} X \arrow[r, "1_X"] \arrow[d, "f"] & X \arrow[r, "1_X"] \arrow[d, "g"] & X \arrow[d, "f"] \\ \mathcal{C} \arrow[r, "k"] & \mathcal{D} \arrow[r, "h"] & \mathcal{C} \end{tikzcd}$$

□

We now proceed to prove theorem 4.1.8.

Proof. If f if a fibration in the J-model structure then among all acyclic cofibration f will have right lifting property w.r.t. the set S since $S \subset \mathbb{JE} \cap Mono$, the class of acyclic

cofibrations. This is exactly the definition of an isofibration, lifting property along inner horns implies that it is an inner fibration where as lifting property along $\{0\} \to J$ is precisely the equivalence lifting condition for isofibration. Conversely suppose that $X \xrightarrow{f} \mathcal{C}$ is an isofibration. The following lifting problem has to be solved where p is a Joyal equivalence and a monomorphism between arbitrary simplicial sets A, B.

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "p"'] & X \arrow[d, "f"] \\ B \arrow[r, "i"'] & \mathcal{C} \end{tikzcd}$$

Apply small object argument to the map i along the set of inner horn inclusions to obtain an extension of the diagram above to the commutative diagram on the right. Since b is an inner fibration and $\mathcal{C}$ is an ∞-category any inner horn to $\mathcal{D}$ can be lifted to $\mathcal{C}$ and then to $\mathcal{D}$ by b. Thus $\mathcal{D}$ is an ∞-category. Since a is inner anodyne it is both a Joyal equivalence and a monomorphism. Since the same is true for p, we find that $a \circ p$ is a monomorphism and a Joyal equivalence. Since $\mathcal{D}$ is an ∞-category , we deduce from theorem 5.1.9 that $a \circ p$ is J-anodyne. But f is an isofibration which means that $f \in inj(S) = inj(J - \text{anodyne})$

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "p"] & X \arrow[dd, "f"] \\ B \arrow[d, "a"] \arrow[ur, dashed] \arrow[dr, "i"] & \\ \mathcal{D} \arrow[r, "b"] & \mathcal{C} \end{tikzcd}$$

Since b is an inner fibration and $\mathcal{C}$ is an ∞-category any inner horn to $\mathcal{D}$ can be lifted to $\mathcal{C}$ and then to $\mathcal{D}$ by b. Thus $\mathcal{D}$ is an ∞-category. Since a is inner anodyne it is both a Joyal equivalence and a monomorphism. Since the same is true for p, we find that $a \circ p$ is a monomorphism and a Joyal equivalence. Since $\mathcal{D}$ is an ∞-category , we deduce from lemma 4.1.9 that $a \circ p$ is J-anodyne. But f is an isofibration which means that $f \in inj(S) = inj(J - \text{anodyne})$, and the lifting problem solves immediately at the corner $\mathcal{D}$ and thus for B via d. □

Corollary 3. *The fibrant objects in the J-model structure are precisely the ∞-categories.*

Proof. If $\mathbf{X}$ is a fibrant object then the map to the terminal object, by the previous theorem,is an isofibration and therefore an inner fibration. Furthermore any map to terminal object satisfies the other condition of isofibration since only equivalence in Δ^0 is the identity. Therefore that any ∞-category is a fibrant object in the J-model structure is clear. □

Theorem 4.1.10. *The J-model structure is left proper.*

Proof. Consider the scenario in proposition 4.1.6, except that now p is just a Joyal equivalence of simplicial sets and i a monomorphism. We have to show that p'^* is a Joyal equivalence. Well, in this case i^* is an isofibration by theorem 3.2.13. And p^* a Joyal equivalence by definition. From proposition 4.1.5 we deduce that p'^* is a Joyal equivalence and the model structure is Left proper. □

We end this section with the following observation that:

Theorem 4.1.11. *The Injective Quillen model structure on simplicial sets is the left Bousfield localization of the J-model structure.*

Proof. Every Joyal equivalence $X \xrightarrow{j} Y$ is a weak equivalence. By definition for any ∞-category $\mathcal{C}$ the induced map $\mathcal{C}^Y \xrightarrow{j^*} \mathcal{C}^X$ is a Joyal equivalence. In particular for any kan complex $\mathcal{K}$ the induced map $\mathcal{K}^Y \xrightarrow{j^*} \mathcal{K}^X$ is a homotopy equivalence of kan complexes. By Gabriel-Zisman lemma we find that j is a weak equivalence. Now the claim is just the definition of left Bousfield localization. □

4.2 Marked model structure

Definition 4.2.1. *A marked simplicial set is a pair $(\mathbf{X}, S)$ where $\mathbf{X}$ is a simplicial set and S is a set of 1-simplices of $\mathbf{X}$ containing all the degenerate ones. We call the morphisms in S as the marked morphisms of S. A map $(\mathbf{X}, S) \xrightarrow{f} (\mathbf{Y}, T)$ of marked simplicial sets is a simplicial map $\mathbf{X} \xrightarrow{f} \mathbf{Y}$ such that $f_1(S) \subseteq T$.*

From now on we will freely write X_S instead of (X, S) to indicate a marked simplicial set. By $X_\sharp$ we will mean the simplicial set whose all edges are marked, for marking all equivalence we write $X_\simeq$ and for marking all the degenerate simplice we write $X_\flat$. The category of marked simplicial sets is Cartesian closed and is denoted sSet_+. The exponentials are given by the corresponding full subcategories on functors which preserve the marked edges.

Definition 4.2.2. *A map $X_S \xrightarrow{f} Y_T$ of marked simplicial sets is called a marked equivalence if for any ∞-category the induced map of ∞-categories on the corresponding full subcategories $\mathcal{C}^{Y_T}_{\simeq} \xrightarrow{f^*} \mathcal{C}^{X_S}_{\simeq}$ is a Joyal equivalence of ∞-categories.*

We do not wish to prove the following theorem but refer to the discussion of J-model structure for a similar argument.

Theorem 4.2.1. ***Marked model structure***
There exists a model structure on marked simplicial sets $sSet_+$ where cofibrations are monomorphisms and weak equivalences are marked equivalences. The fibrant objects are ∞-categories with all of the equivalences marked.

Now we state a different proof of the existence of localization for ∞-categories.

Theorem 4.2.2. *A localization of an ∞-category $\mathcal{C}$ along any set S of edges always exists and is given by a fibrant replacement of $\mathcal{C}_S$ in marked model structure.*

Proof. Let R be a fibrant replacement functor in the marked model structure. Then By definition we have a marked equivalence $\mathcal{C}_S \to R(\mathcal{C}_S)$, which translates to saying that for any ∞-category $\mathcal{E}$ the induces map $\mathcal{E}^{R(\mathcal{C}_S)}_{\simeq} \to \mathcal{E}^{\mathcal{C}_S}_{\simeq}$ is a Joyal equivalence. But this is precisely the map $\mathcal{E}^{R(\mathcal{C}_S)} \to \mathcal{E}^{\mathcal{C}}_{S}$ since the marked edges of $R(\mathcal{C}_S)$ are equivalences only. This implies that $R(\mathcal{C}_S)$ is a Dwyer Kan localization of $\mathcal{C}$ along S. $\square$

4.3 strengthening and un-strengthening construction

In this section we will state Lurie's idea of strengthening unstrengthening method, which is an ∞ categorical generalization of the idea of (due to Grothendieck) equivalence of fibered categories over a base and the category of $textCat$ valued pseudo-functors on that base. Cartesian and Co-cartesian morphisms in ordinary fibered categories, will be replaced by the following definition (similar to some extent). From now on all morphisms are going to be inner fibrations.

Definition 4.3.1. *Let $X \xrightarrow{p} Y$ be an inner fibration of simplicial sets. An edge $x \xrightarrow{f} y$ in X is called $p-$ cartesian (dually p-cocartesian) if any lifting problem on the left (respectively*

right) can be solved.

$$\begin{tikzcd} \Delta^1 \arrow[r, "\{n-1,n\}"'] \arrow[rr, bend left, "f"] & \Lambda^n_n \arrow[r] \arrow[d] & X \arrow[d, "p"] \\ & \Delta^n \arrow[r] \arrow[ur, dashed] & Y \end{tikzcd} \qquad \begin{tikzcd} \Delta^1 \arrow[r, "\{0,1\}"'] \arrow[rr, bend left, "f"] & \Lambda^n_0 \arrow[r] \arrow[d] & X \arrow[d, "p"] \\ & \Delta^n \arrow[r] \arrow[ur, dashed] & Y \end{tikzcd} \tag{4.4}$$

or equivalently, (since $\partial\Delta^{n-2} * \Delta^0 \coprod_{\partial\Delta^{n-2}*\{0\}} \Delta^{n-2} * \{0\} \longrightarrow \Delta^{n-2} * \Delta^0$ *is actually the outer horn inclusion* $\Lambda^n_n \hookrightarrow \Delta^n$ *and its dual statement) due to lemma 2.4.3 (and its dual) if* $X_{/f} \to X_{/y} \times_{Y_{/p(y)}} Y_{/pf}$ *(dually* $X_{f/} \to X_{y/} \times_{Y_{p(y)/}} Y_{pf/}$*) is a trivial fibration. By* $X^p_\natural$ *we indicate the simplicial set with all of its p-cartesian maps marked.*

We also have a replacement for the notion of Grothendieck fibration and Grothendieck co fibration.

Definition 4.3.2. *An inner fibration* $X \xrightarrow{p} Y$ *of simplicial sets is a cartesian fibration(dually cocartesian fibration) if the left (respectively right) diagram is solvable by a p-cartesian lift.*

$$\begin{tikzcd} \Delta^0 \arrow[r] \arrow[d, "\{1\}"'] & X \arrow[d, "p"] \\ \Delta^1 \arrow[r] \arrow[ur, dashed] & Y \end{tikzcd} \qquad \begin{tikzcd} \Delta^0 \arrow[r] \arrow[d, "\{0\}"'] & X \arrow[d, "p"] \\ \Delta^1 \arrow[r] \arrow[ur, dashed] & Y \end{tikzcd} \tag{4.5}$$

Now consider full ∞subcategory of $Cat_{\infty/\mathcal{B}}$ on objects which are cartesian fibration and on one simplices given by ∞-functors which preserve cartesian arrows and denote it by $Cart(\mathcal{B})$. This is an exact replacement of the notion of the category of fibered categories. And on the other hand we have the ∞-category $Cat_\infty^{\mathcal{B}^{op}}$ which is a replacement for the notion of Cat valued pseudo-functors on an ordinary category. The following result by Lurie, is an ∞-categorical replacement of Grothendieck equivalence for ordinary categories.

Theorem 4.3.1. *There is Joyal equivalence between the two* ∞*-categories as mentioned above,* $Cart(\mathcal{B}) \simeq Cat_\infty^{\mathcal{B}^{op}}$*. Dually there is a Joyal equivalence* $CoCart(\mathcal{B}) \simeq Cat_\infty^{\mathcal{B}}$*.*

We won't give a detailed proof, but will briefly mention this on 0-simplices and 1-simplices. Given a cocartesian fibration $\mathcal{E} \xrightarrow{p} \mathcal{B}$ over an ∞-category $\mathcal{B}$ send it to the

∞-functor $\mathcal{B} \xrightarrow{p_{\mathcal{E}}} Cat_\infty$ such that $p_{\mathcal{E}}(x) = \mathcal{E}_x = \mathcal{E} \times_{\mathcal{B}} \{x\}$ for $x \in \mathcal{B}$. For a 1-simplex $x \xrightarrow{f} y$ in $\mathcal{B}$ the map $\mathcal{E}_x \xrightarrow{p_{\mathcal{B}}(f)} \mathcal{E}_y$ of ∞-categories is given as follows. Consider the pullback :

$$\begin{tikzcd} A \arrow[r] \arrow[d] & \mathcal{E}_\natural^{\Delta^1_\sharp} \arrow[d, "p^*"] \\ \Delta^0 \arrow[r, "f"] & \mathcal{B}^{\Delta^1} \end{tikzcd}$$

We have canonical maps $A \to \mathcal{E}_x$ and $A \to \mathcal{E}_y$ where the first one is a trivial fibration. Choose a section of the first map and consider the composite $\mathcal{E}_x \to A\mathcal{E}_y$ which is the map we wanted. The higher simplices are mapped canonically.

Clearly every left fibration is cocartesian and every right fibration is cartesian. We get the following restriction of the above equivalence:

Theorem 4.3.2. *The above equivalence restricts to $Rfb(\mathcal{B}) \simeq Spc^{\mathcal{B}^{op}}$ and dually $Lfb(\mathcal{B}) \simeq Spc^{\mathcal{B}}$ on left and right fibration.*

Chapter 5

Further categorical structures

All concepts existing in ordinary category theory can be easily translated to the language of ∞-categories. We will briefly mention them here to point out the richness of the theory, without going to details or proofs. Nothing is new in this chapter.

5.1 The Yoneda lemma

Given an object x in an ∞-category $\mathcal{C}$ we have a left fibration $\mathcal{C}_{x/} \to \mathcal{C}$, which by strengthening un-strengthening construction correspond to an ∞-functor $\mathcal{C} \to Spc$ which on each object $y \in \mathcal{C}$ is given by the ∞hom $\mathcal{C}_1^\infty(x, y)$. We denote this ∞-functor by $\mathcal{C}_1^\infty(x, -)$. Dually we have $\mathcal{C}^{op} \xrightarrow{\mathcal{C}_1^\infty(-,x)} Spc$.

Theorem 5.1.1. ***The ∞ Yoneda lemma:***
For any ∞-functors $\mathcal{C} \xrightarrow{F} Spc$ and $\mathcal{C}^{op} \xrightarrow{G} Spc$, there are natural Joyal equivalences of ∞-categories (in fact, a homotopy equivalence of kan complexes) as follows:

$$(Spc^{\mathcal{C}})_1^\infty\left(\mathcal{C}_1^\infty(x, -), F\right) \simeq F(x) \tag{5.1}$$

$$(Spc^{\mathcal{C}^{op}})_1^\infty\left(\mathcal{C}_1^\infty(-, x), G\right) \simeq G(x) \tag{5.2}$$

5.2 Limits and Colimits

Definition 5.2.1. *An objects x in an ∞-category is an initial object if for any other object $y \in \mathcal{C}$ the ∞hom $\mathcal{C}_1^\infty(x, y)$ is contractible. Dually, x is terminal if for any other object $y \in \mathcal{C}$ $\mathcal{C}_1^\infty(y, x)$ is contractible.*

Definition 5.2.2. *Let $F : B \to \mathcal{C}$ be a simplicial map where $\mathcal{C}$ is an ∞-category and B is an arbitrary simplicial set. We say that F has a limit (dually a colimit) in $\mathcal{C}$ if $\mathcal{C}_{/F}$ (dually $\mathcal{C}_{F/}$) has a terminal (dually initial) object. The image of that object under the corresponding forgetful functor is called the limit object (dually the colimit object) of F*

Results about limits and colimits are similar to the case in ordinary category theory.

Theorem 5.2.1. *An ∞-category has all limits (dually colimits) iff it has small products (limits indexed by 0-dimensional simplicial set) and pull back (limits indexed by left 2 horn) exists (dually coproducts and pushouts.)*

Definition 5.2.3. *A map of simplicial set $K \xrightarrow{f} L$ is called cofinal if for all right fibration $X \xrightarrow{p} L$ the induced map $p_*^{-1}(1_L) \to p_*^{-1}(f)$ on fibers is a Joyal equivalence.*

Theorem 5.2.2. *Let $K \xrightarrow{p} L$ and $\mathcal{L} \xrightarrow{f} \mathcal{C}$ be diagrams where $\mathcal{C}$ is an ∞-category. If p is cofinal then fp has a colimit iff f has a colimit and in all such situation the colimits agree.*

Theorem 5.2.3. ***Quillen's theorem A:***
Let $D \xrightarrow{p} \mathcal{C}$ be a diagram where $\mathcal{C}$ is an ∞-category and D is a simplicial set. Then p is cofinal iff for each object $c \in \mathcal{C}$ the simplicial set $D_{c/}$ is weakly contractible.

5.3 Adjunction

Definition 5.3.1. *An adjunction is a bicartesian (i.e. both cartesian and cocartesian) functor $\mathcal{E} \xrightarrow{p} \Delta^1$.*

Note that by strengthening and un-strengthening of equivalences we get two maps $\mathcal{E}_0 \xrightarrow{f} \mathcal{E}_1$ and $\mathcal{E}_1 \xrightarrow{g} \mathcal{E}_0$ which we call the adjoint maps corresponding to the adjunction. In fact just as the case of ordinary categories we find a Joyal equivalence of ∞homs: $\mathcal{E}_{\infty 1}^\infty(f(x), y) \simeq$

$\mathcal{E}_{\prime 1}^{\infty}(x, gy)$, indicating that f is a left adjoint to g and g is a right adjoint to f. We have the following version of Fried's adjoint functor theorem for∞-categories

Theorem 5.3.1. *Let $\mathcal{C}$ be a locally small ∞-category containing an essentially small colimit dense full subcategory and let $\mathcal{D}$ another locally small ∞-category. An ∞-functor $\mathcal{C} \xrightarrow{F} \mathcal{D}$ admits a right adjoint iff F preserves colimits.*

By colimit dense subcategory we mean a subcategory such that every object in the bigger category is a colimit of some diagram in the subcategory. And by locally small we mean that the ∞homs are essentially small that is Joyal equivalent to a small simplicial set. And by small we mean small in the fixed universe of our choice.

Chapter 6

Conclusion

The theory that we have presented here is a generalization of our existing knowledge of ordinary categories. They enjoy the same rich structures as ordinary categories, in fact, it does even more, for being reluctant to the idea of isomorphisms, and are thus stronger in nature. Most of the existing knowledge in mathematics can be hoped to have a natural generalization to the language of ∞-categories. One such generalisation has been given by Lurie in his Higher topos theory. Other generalizations include Higher order logic and stable homotopy theory, also due to Lurie. Derived algebraic geometry is one of the first successful application of the theory. Some recent development in Topology or more specifically in cyclic homology theory, topological modular forms has given us entirely new view towards the subjects. We wish to find better suited version for homotopy type theory and higher algebraic theories including enriched ∞-category theory in future. To conclude the story we present the following diagram comparing basic category theory with ∞-category theory.

$$\begin{tikzcd}
 & Gpd \arrow[r, hook] \arrow[dd, dotted, "N"] & Cat \arrow[dr, hook, "N"] \arrow[dd, dotted, "N"] & \\
Sets \arrow[ur, hook] \arrow[dr, hook] & & & sSets \\
 & \infty Gpd \arrow[r, hook] & \infty Cat \arrow[ur, hook] &
\end{tikzcd}$$

This is however just a slice of what is called an ∞Cosmos, other leaves would involve other notions of $(\infty, 1)$-categories.

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